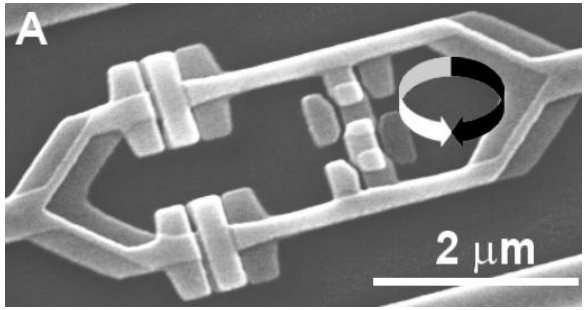
You Did Not Just Read This

or did you?

SquInT: Quantum Algorithms

188198812920607963838697239461650439807163563379417382700763356422988859715234665485319060606504743045317388011303396716199692321205734031879550656996221305168759307650257059





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⊗

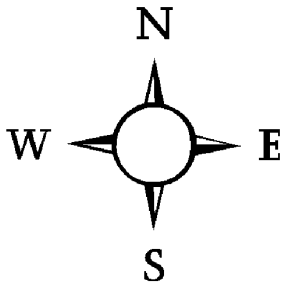
4727721461074353025362230719730482246329146953020971164598521711305207112563635903975 27

Dave Bacon

University of Washington

Department of Computer Science & Engineering

Southwest





Southwest Airlines Route Map



Today

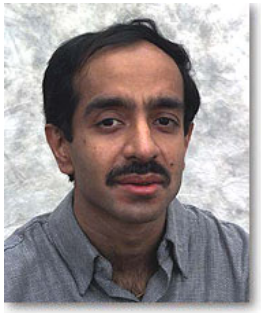
Two 1 Hour Lectures

(you poor souls)

Quantum Algorithms
of the Shor Kind



Quantum Algorithms
of the Grover Kind



Quantum Algorithms
of the Future Kind

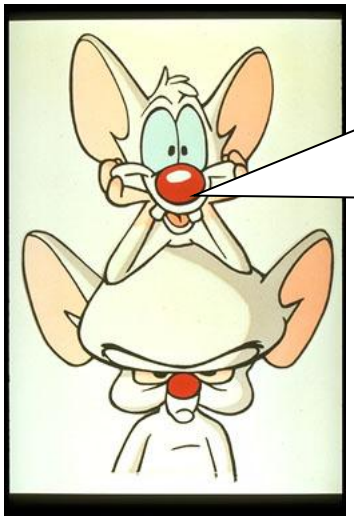


your face here?

Quantum Algorithms



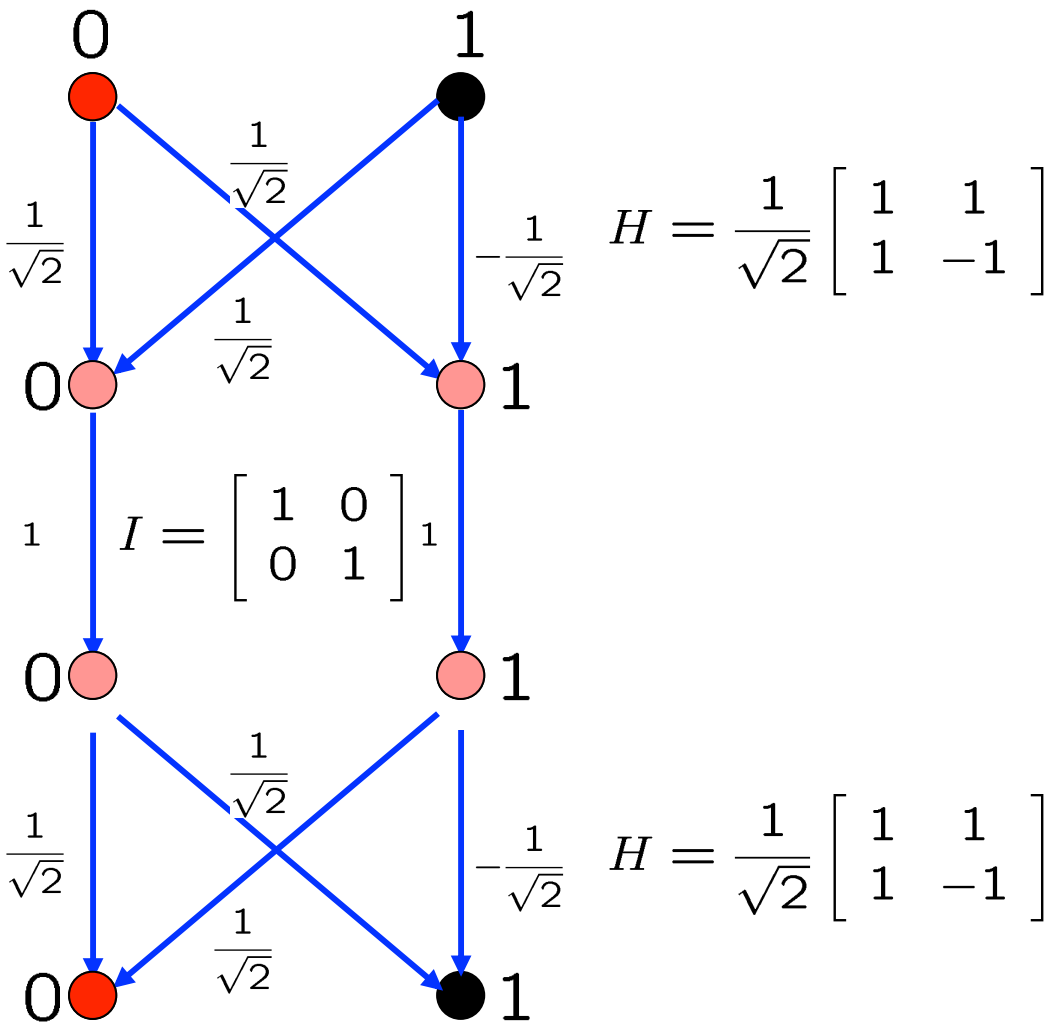
"If I had a good idea in quantum algorithms that's what I'd be working on" – John Preskill, 2004

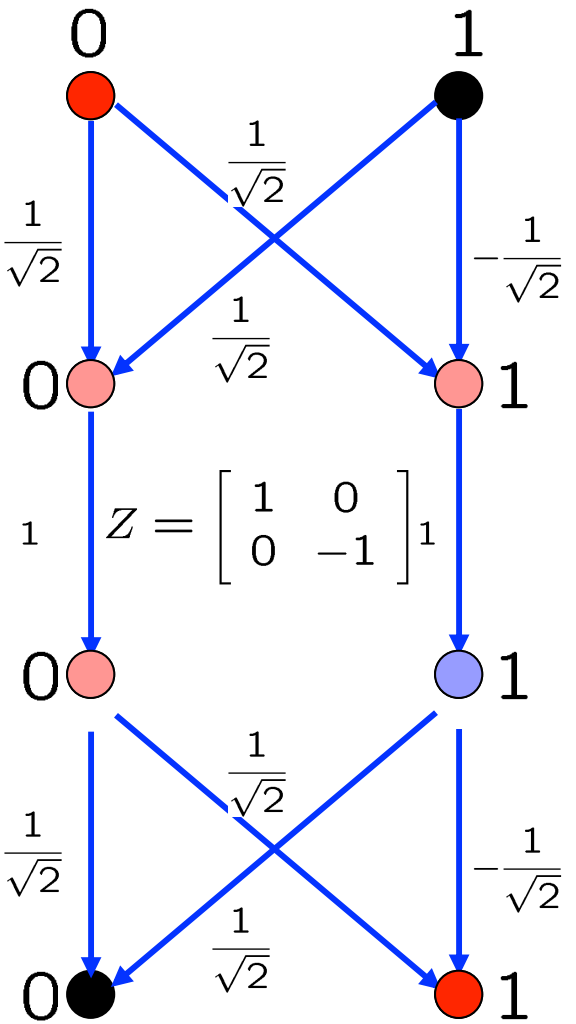


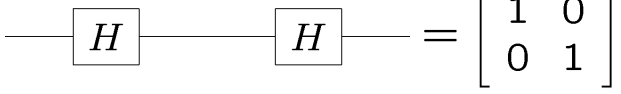
What happens to quantum computing if no significant new quantum algorithms are discovered?

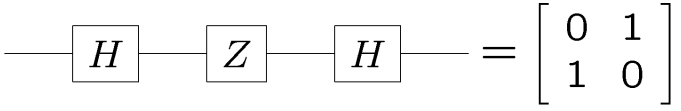
Most important, most exciting, most frustrating.

Interference

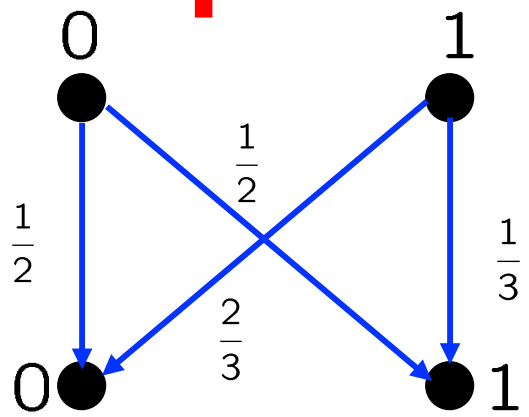








Compare To…



probabilistic classical evolution of a finite state machine

All of the transitions are labeled by **postive** numbers:
hence no interference

Note we can simulate interference on a classical system, but we know of no efficient way to simulate quantum interference of a quantum system with the same resource size as that quantum system.

Can we use interference for algorithms?

In the Beginning There Was…



David
Deutsch



Dr. Falken
(from the movie War Games)

David Speaks



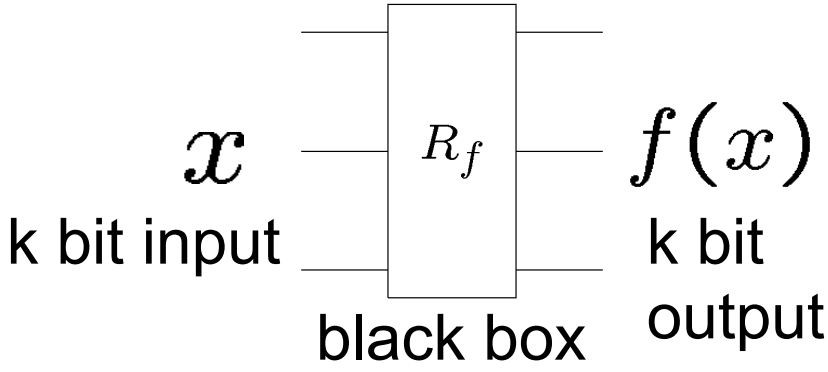
David Deutsch 1985

"Complexity theory has been mainly concerned with constraints upon the computation of functions: which functions can be computed, how fast, and with use of how much memory. With quantum computers, as with classical stochastic computers, one must also ask 'and with what probability?' We have seen that the minimum computation time for certain tasks can be lower for Q than for T . Complexity theory for Q deserves further investigation."

Q = quantum computers
T = classical computers

Classical Promise Problem Query Complexity

Given: A black box which computes some function



Promise: the function belongs to a set $\mathcal{S}$ which is a subset of all possible functions.

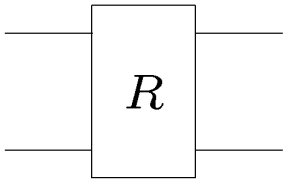
Properties: the set $\mathcal{S}$ can be divided into disjoint subsets $\mathcal{S}_1, \mathcal{S}_2, \ldots, \mathcal{S}_m$

Problem: What is the minimal number of times we have to use (query) the black box in order to determine which subset $\mathcal{S}_i$ the function belongs to?

Deutsch's Problem (Revisionist)

Suppose you are given a black box which computes one of the following four reversible classical gates:

2 bits input



2 bits output

$00 \to 00$	$00 \to 01$	$00 \to 00$	$00 \to 01$
$01 \to 01$	$01 \to 00$	$01 \to 01$	$01 \to 00$
$10 \to 10$	$10 \to 11$	$10 \to 11$	$10 \to 10$
$11 \to 11$	$11 \to 10$	$11 \to 10$	$11 \to 11$
"identity"	NOT 2nd bit	controlled-NOT	controlled-NOT + NOT 2nd bit
	$\mathcal{S}_1$		$\mathcal{S}_2$

Deutsch's (Classical) Problem: What is the minimal number of times we have to use this black box to determine whether we are given one of the first two or the second two functions?

Deutsch's Problem (Revisionist)

$00 \to 00$	$00 \to 01$	$00 \to 00$	$00 \to 01$
$01 \to 01$	$01 \to 00$	$01 \to 01$	$01 \to 00$
$10 \to 10$	$10 \to 11$	$10 \to 11$	$10 \to 10$
$11 \to 11$	$11 \to 10$	$11 \to 10$	$11 \to 11$
"identity"	NOT 2nd bit	controlled-NOT	controlled-NOT + NOT 2nd bit
$\mathcal{S}_1$		$\mathcal{S}_2$	

Deutsch's (Classical) Problem: What is the minimal number of times we have to use this black box to determine whether we are given one of the first two or the second two functions?

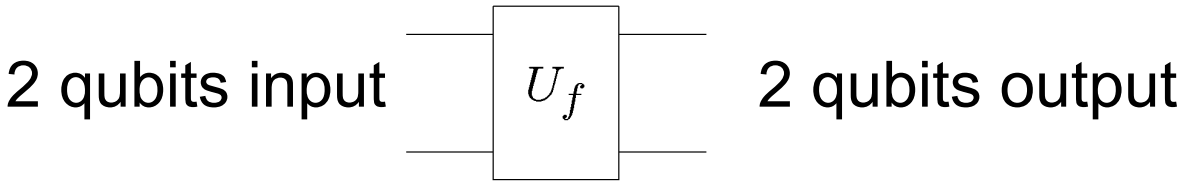
1. It is necessary to query more than once.
2. Two queries are sufficient.

"classical query complexity of Deutsch's problem is two"

Quantum Deutsch's Problem

$00 \rightarrow 00$	$00 \rightarrow 01$	$00 \rightarrow 00$	$00 \rightarrow 01$
$01 \rightarrow 01$	$01 \rightarrow 00$	$01 \rightarrow 01$	$01 \rightarrow 00$
$10 \rightarrow 10$	$10 \rightarrow 11$	$10 \rightarrow 11$	$10 \rightarrow 10$
$11 \rightarrow 11$	$11 \rightarrow 10$	$11 \rightarrow 10$	$11 \rightarrow 11$
"identity"	NOT 2nd bit	controlled-NOT	controlled-NOT + NOT 2nd bit
$\mathcal{S}_1$		$\mathcal{S}_2$	

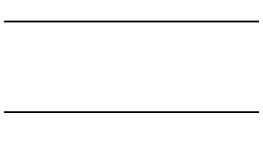
These functions are all reversible (one-to-one). Thus we can implement them on a quantum computer as reversible quantum gates.



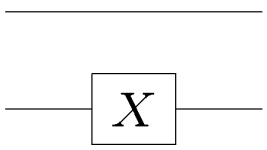
Classical to Quantum Deutsch

$00 \rightarrow 00$	$00 \rightarrow 01$	$00 \rightarrow 00$	$00 \rightarrow 01$
$01 \rightarrow 01$	$01 \rightarrow 00$	$01 \rightarrow 01$	$01 \rightarrow 00$
$10 \rightarrow 10$	$10 \rightarrow 11$	$10 \rightarrow 11$	$10 \rightarrow 10$
$11 \rightarrow 11$	$11 \rightarrow 10$	$11 \rightarrow 10$	$11 \rightarrow 11$
"identity"	NOT 2nd bit	controlled-NOT	controlled-NOT + NOT 2nd bit

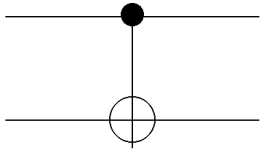
convert to quantum gates



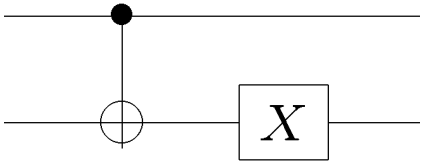
$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$



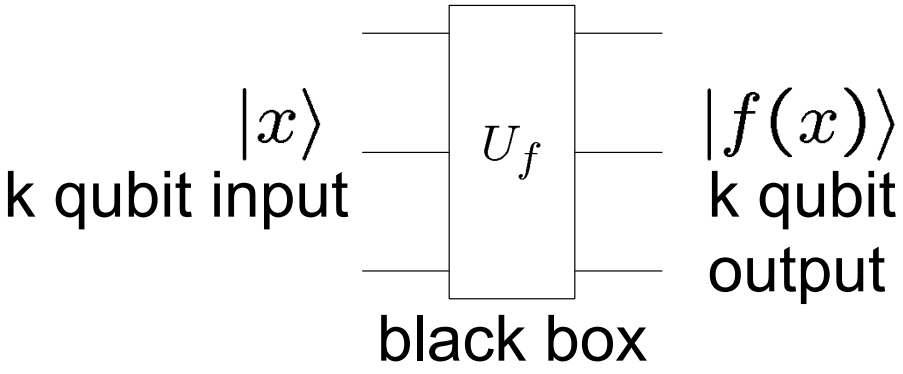
$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$



$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Quantum Promise Query Complexity

Given: A quantum gate which, when used as a classical device computes a reversible function



Promise: the function belongs to a set $\mathcal{S}$ which is a subset of all possible functions.

Properties: the set $\mathcal{S}$ can be divided into disjoint subsets $\mathcal{S}_1, \mathcal{S}_2, \ldots, \mathcal{S}_m$

Problem: What is the minimal number of times we have to use (query) the quantum gate in order to determine which subset $\mathcal{S}_i$ the function belongs to?

Fun With Quantum Circuits

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$ "bit flip"

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$ "phase flip"

$$H = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

"Hadamard"

$$H \; H = I$$

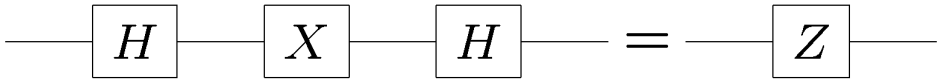
$$H \; X \; H = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

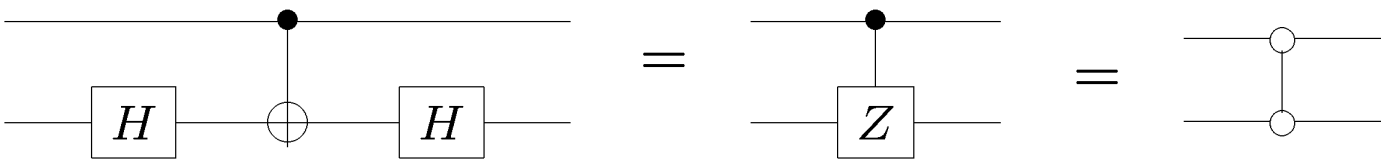
$$= \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

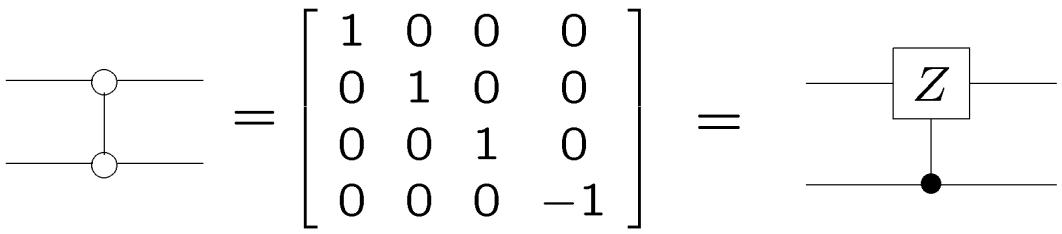
$$= \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = Z$$

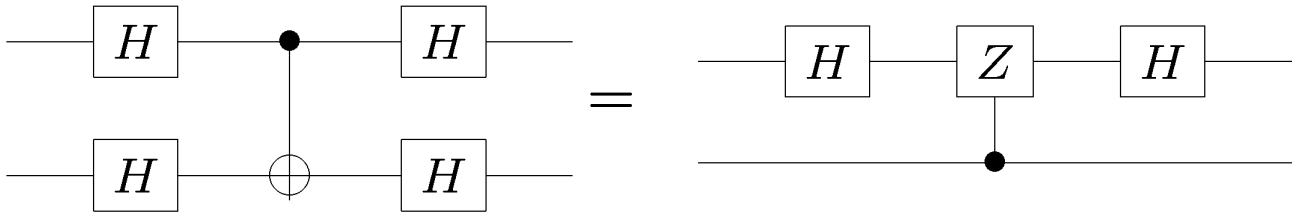
The Hadamard exchanges the role of bit flips and phase flips

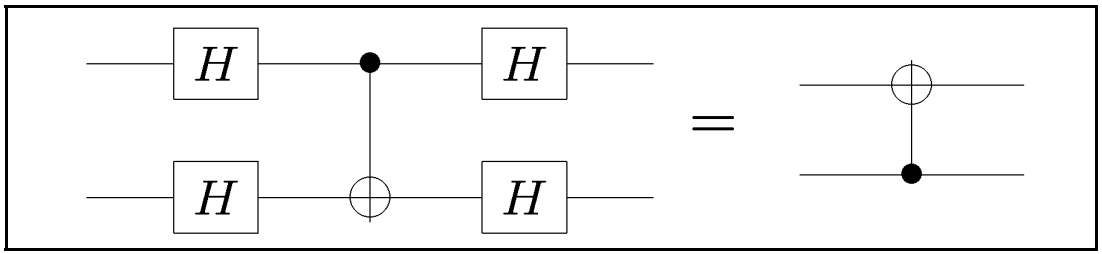
More Fun With Quantum Circuits

Using 



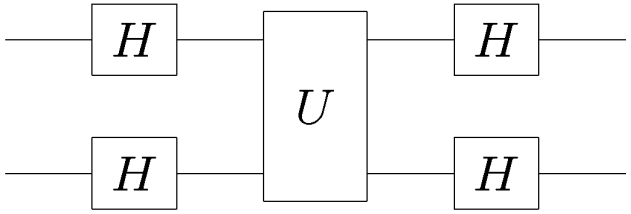


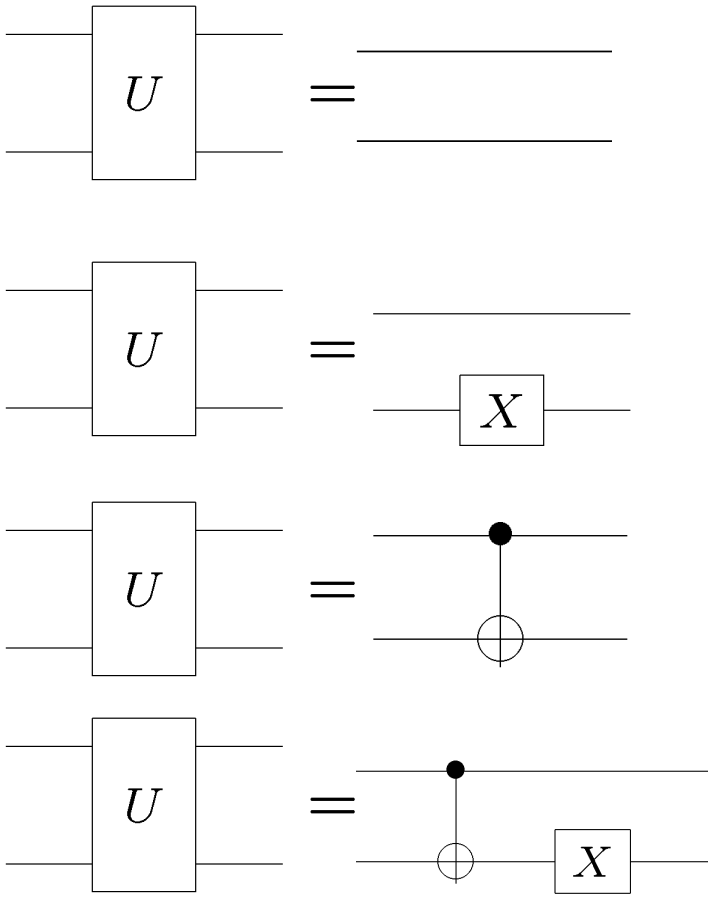


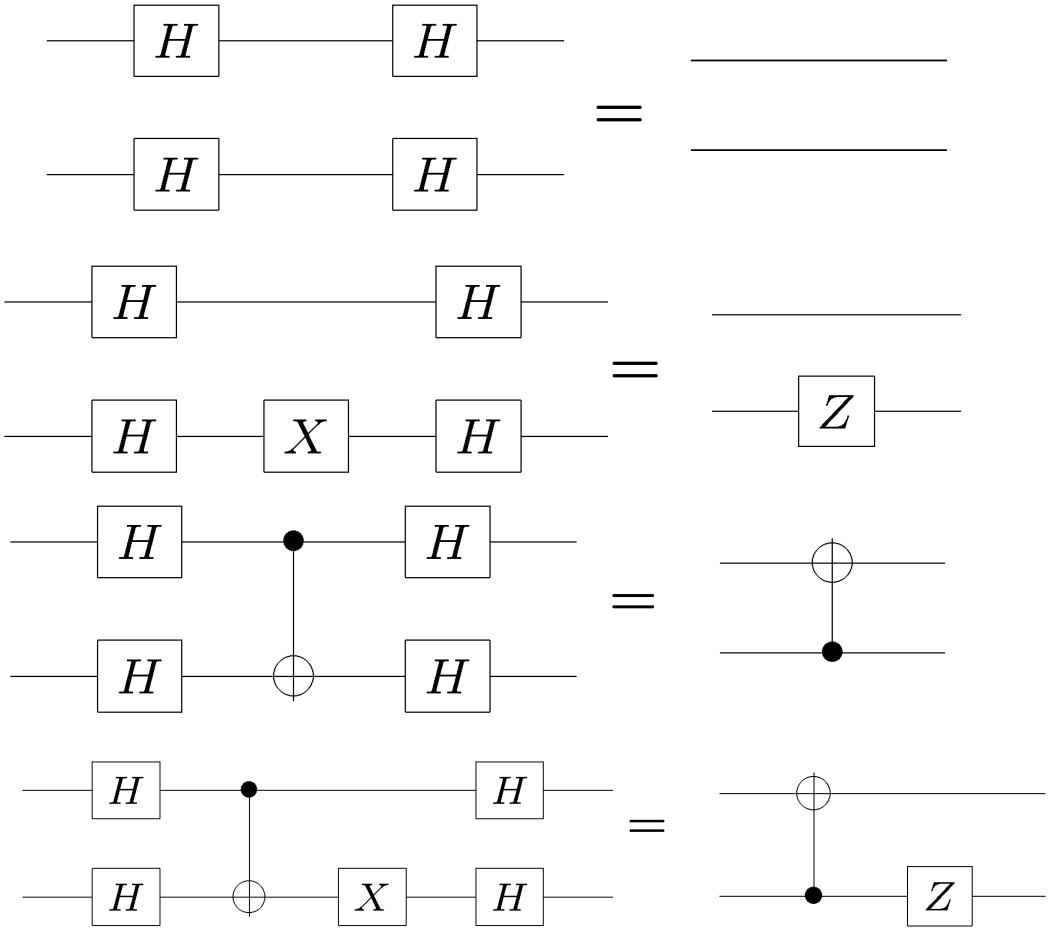


Quantum Deutsch

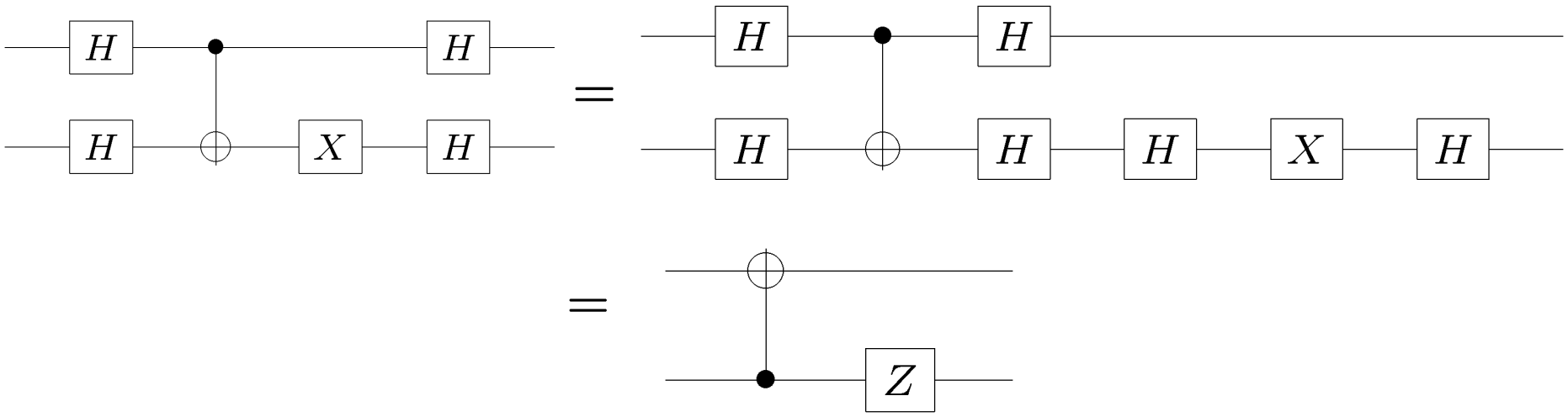
What if we perform Hadamards before and after the quantum gate:



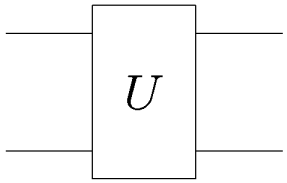


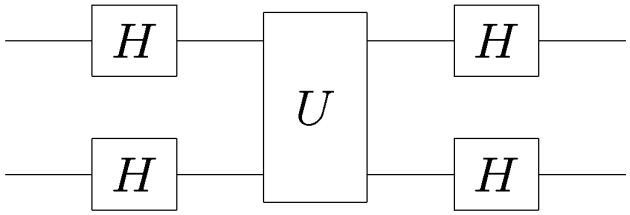


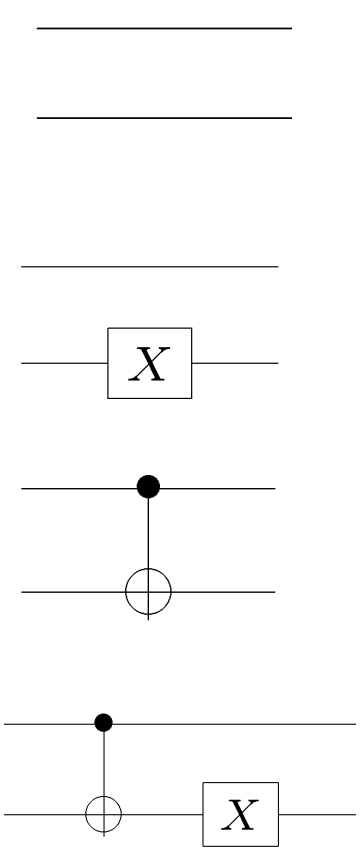
That Last One

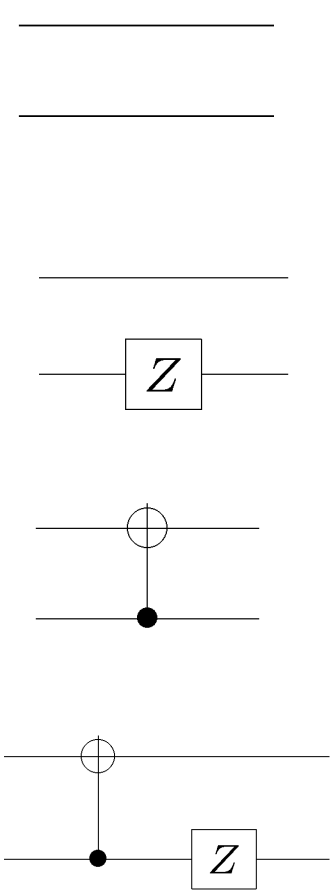


Agai
n

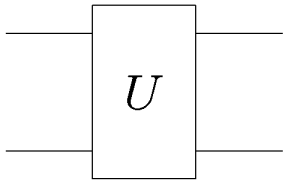


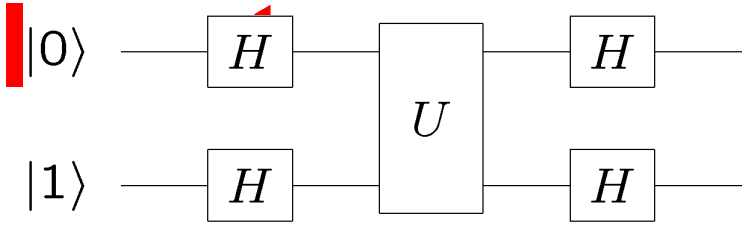


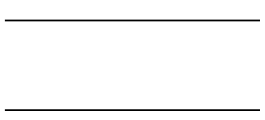


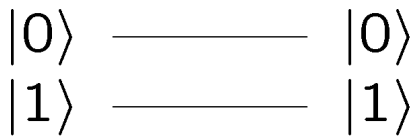


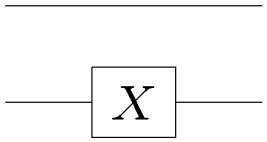
Some

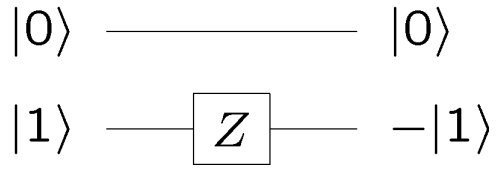


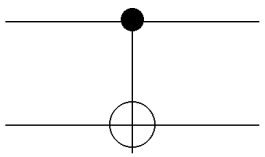


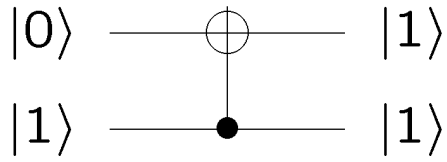


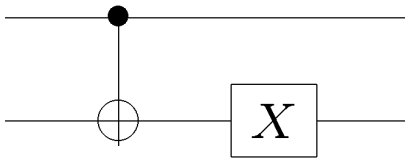


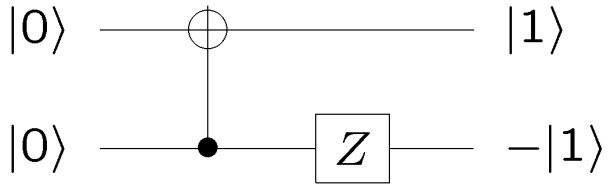




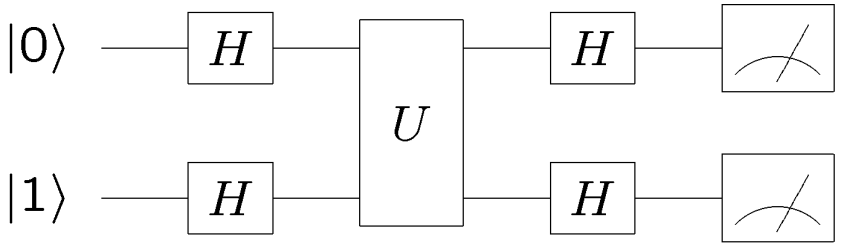


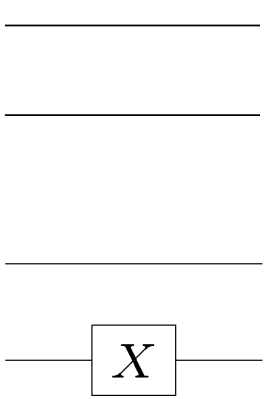


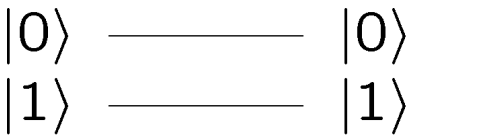


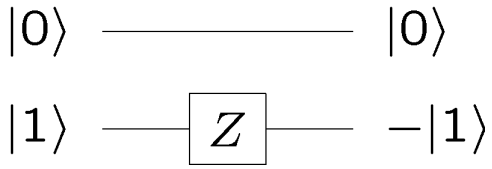


Quantum Deutsch

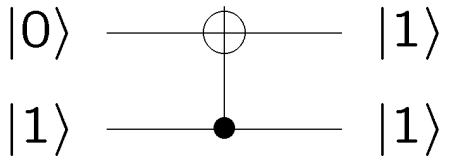


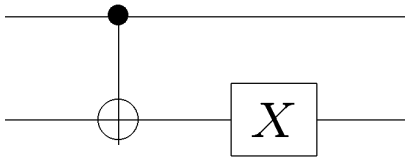


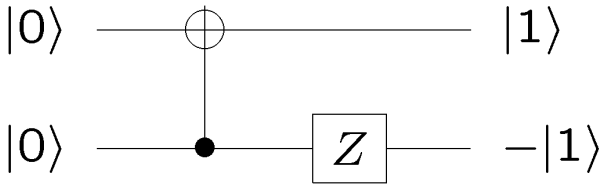




= 0

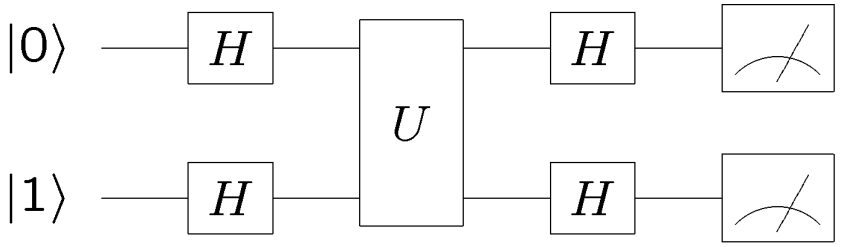






= 1

Quantum Deutsch

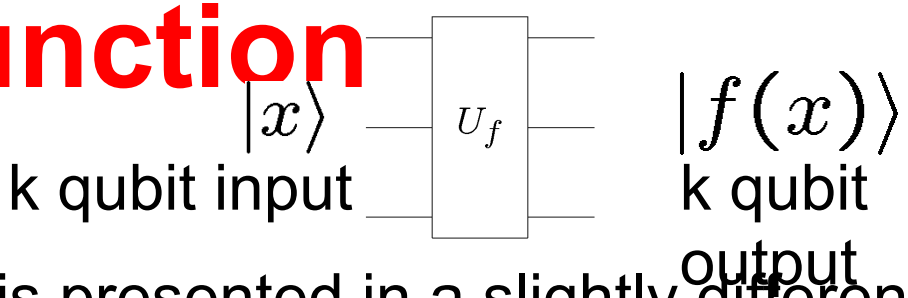


By querying with quantum states we are able to distinguish the first two (constant) from the second two (balanced) with
only one use of the quantum gate!

Two uses of the classical gates
Versus
One use of the quantum gate

quantum speedup

An Aside on Function

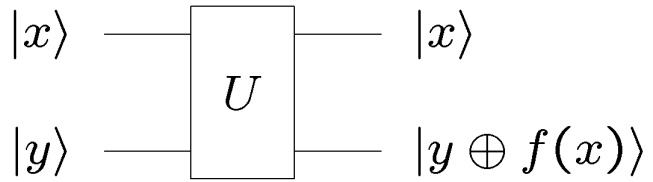


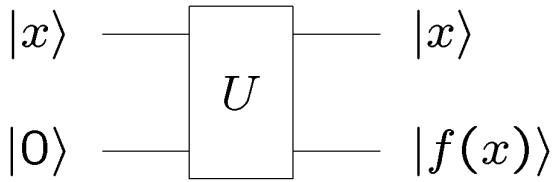
Normally this is presented in a slightly different manner. The reason for this has to do with how we calculate non-reversible functions on a quantum computer

Suppose for example you want to implement the function

$$f(x) = 0 \qquad x = \{0, 1\}$$

This is not a reversible function, but we can implement it as





exclusive or: $a \oplus b = \begin{cases} 0 \text{ if} a = b \\ 1 \text{ otherwise} \end{cases}$

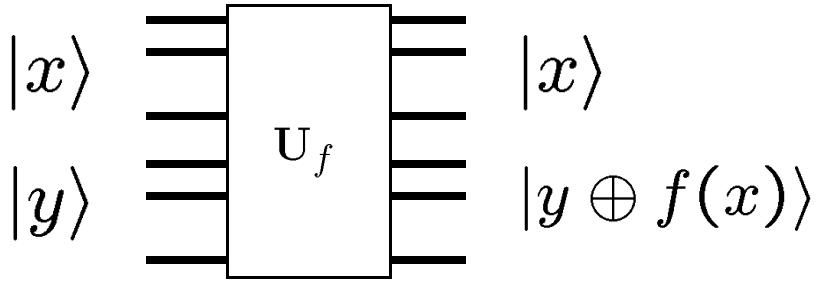
An Aside on Functions

Generically we can compute a non-reversible function using

function from n bits to k bits:

$$f : x \in \{0,1\}^n \rightarrow \{0,1\}^k$$

n qubits
k qubits

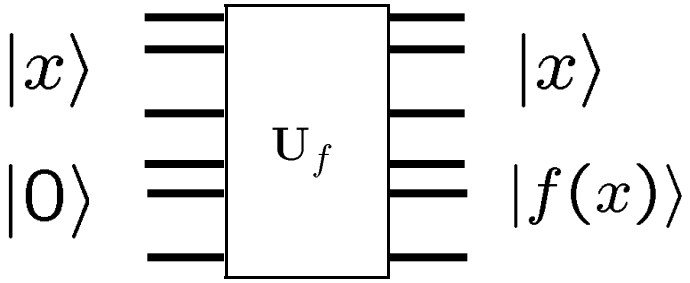


$$y, f(x) \in \{0,1\}^k$$

$$x \in \{0,1\}^n$$

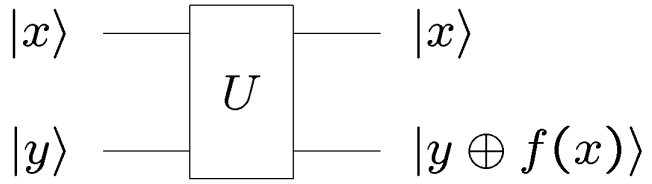
$y \oplus f(x)$ is a bitwise exclusive or

Such that, with proper input we can calculate f:



From This Perspective

$00 \to 00$	$00 \to 01$	$00 \to 00$	$00 \to 01$
$01 \to 01$	$01 \to 00$	$01 \to 01$	$01 \to 00$
$10 \to 10$	$10 \to 11$	$10 \to 11$	$10 \to 10$
$11 \to 11$	$11 \to 10$	$11 \to 10$	$11 \to 11$
"identity"	NOT 2nd bit	controlled-NOT	controlled-NOT + NOT 2nd bit

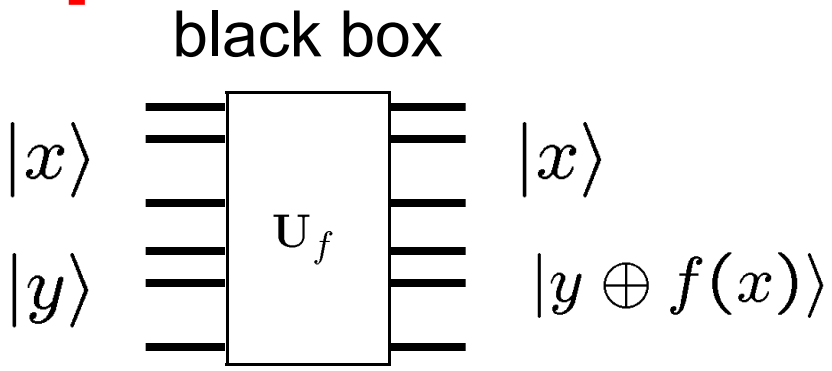


$f(x) = 0$	$f(x) = 1$	$f(x) = x$	$f(x) = \bar{x}$
constant functions		balanced functions	

Deutsch's problem is to distinguish constant from balanced

Query Complexities

$$f : x \in \{0,1\}^n \rightarrow \{0,1\}^k$$

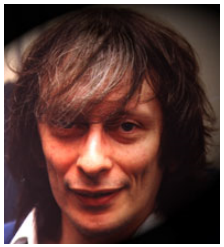


	probability of failure
Exact classical query complexity	0
Bounded error classical query complexity	$< \frac{1}{2}$
Exact quantum query complexity	0
Bounded error quantum query complexity	$< \frac{1}{2}$

Bounded error algorithms are allowed to fail with a bounded probability of failure.

Random Extra Slide

Quantum Algorithms



David Deutsch



Richard Jozsa

1992: Deutsch-Jozsa Algorithm

Exact classical q. complexity: $2^{n-1}+1$

Bounded error classical q. complexity: $O(1)$

Exact quantum q. complexity: 1



Umesh Vazirani



Ethan Bernstein

1993: Bernstein-Vazirani Algorithm (non-recursive)

Exact classical q. complexity: n

Bounded error classical q. complexity: $\Omega(n)$

Exact quantum q. complexity: 1

Quantum Algorithms



Umesh Vazirani



Ethan Bernstein

1993: Bernstein-Vazirani Algorithm (recursive)

Bounded error classical q. complexity: $n^{\Omega(\log n)}$

Exact quantum q. complexity: $O(n \log n)$

(first super-polynomial separation)



Dan Simon

1994: Simon's Algorithm

Bounded error classical q. complexity: $\Omega(2^{n/2})$

Bounded error quantum q. complexity: $O(n)$

(first exponential separation)

Generalizing Simon's algorithm, in 1994, Peter Shor was able to show that…

The Factoring Firestorm



Peter Shor 1994

188198812920607963383697239461650439807163563379417382700763356422988859715234665485319060606504743045317388011303396716199623212057340318795506569622130516875930765025705 9

=

3980750864240649373971255005503864911990643623425267084063851895759463889572617685833172727214610743530253622307197304822463291469530209711645985217113052071125636359039752 7

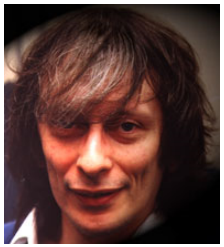
×

Best classical algorithm takes time $O(\exp(n^{1/3}))$

Shor's quantum algorithm takes time $O(n^2 \log n)$

An efficient algorithm for factoring **breaks** the RSA public key cryptosystem

The Golden Years



David Deutsc



Richar d

1992: Deutsch-Jozsa Algorithm



Umesh Vazirani



Ethan Bernstein

1993: Bernstein-Vazirani Algorithm (non-recursive)



Dan Simon

1994: Simon's Algorithm



Peter Shor

1994: Shor's Algorithm

Deutsch-Jozsa Problem

Given: A function with n bit strings as input and one bit as output

$$f : x \in \{0,1\}^n \rightarrow \{0,1\}$$

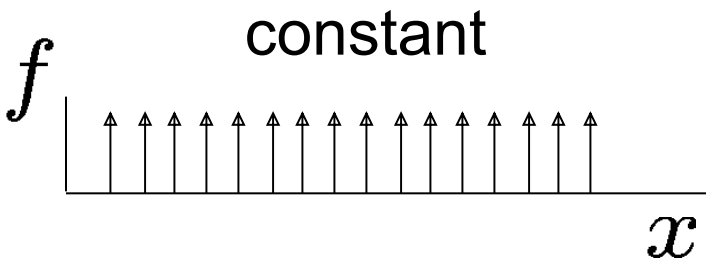
(this will be a non-reversible function)

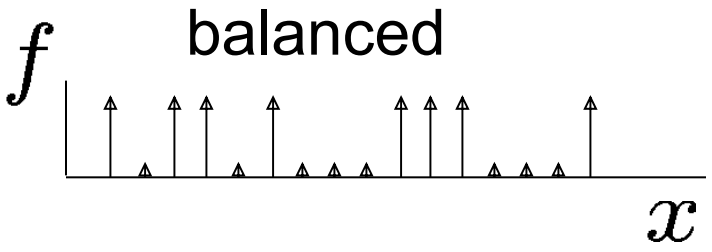
Promise: The function is either constant or balance.

constant function: $\forall x \in \{0,1\}^n, f(x) = b$

balanced function: $\forall x \in \mathcal{H}, f(x) = 0, \forall x \notin \mathcal{H}, f(x) = 1$

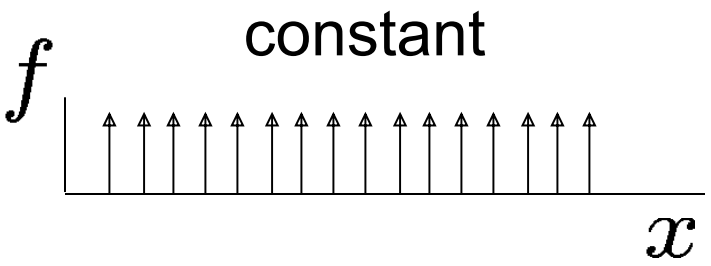
$$\mathcal{H} \subset \{0,1\}^n, |\mathcal{H}| = 2^{n-1}$$

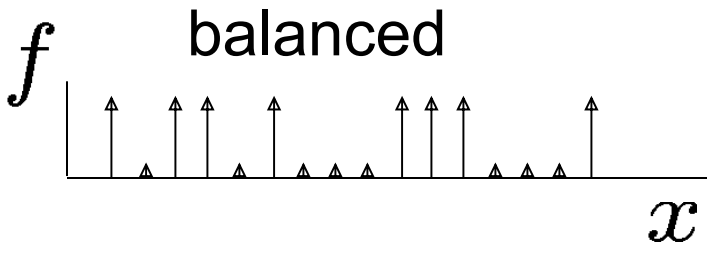




Problem: determine whether the function is constant or balanced.

Classical Deutsch-Jozsa



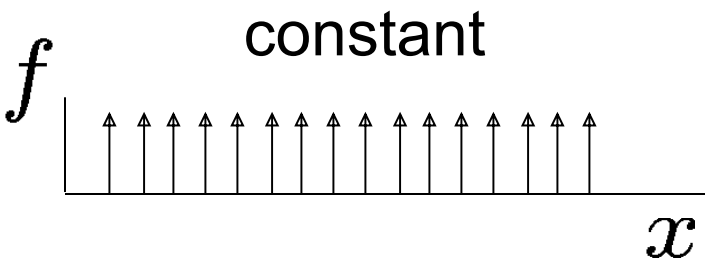


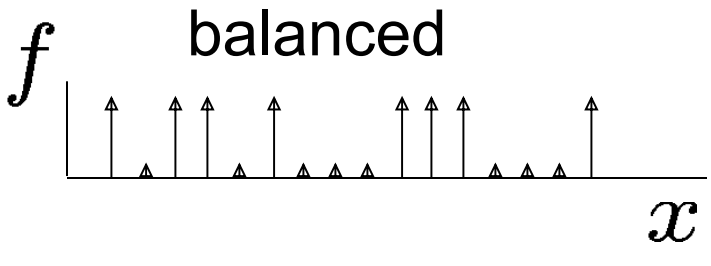
Problem: determine whether the function is constant or balanced.

No failure allowed: we need to query in the worst case $2^{n-1}+1$ values of f to distinguish between constant and balanced

Exact classical q. complexity: $2^{n-1}+1$

Classical Deutsch-Jozsa





Problem: determine whether the function is constant or balanced.

Bounded error: Query two different random values of the function.

If they are equal, guess constant.
Otherwise, guess balanced.

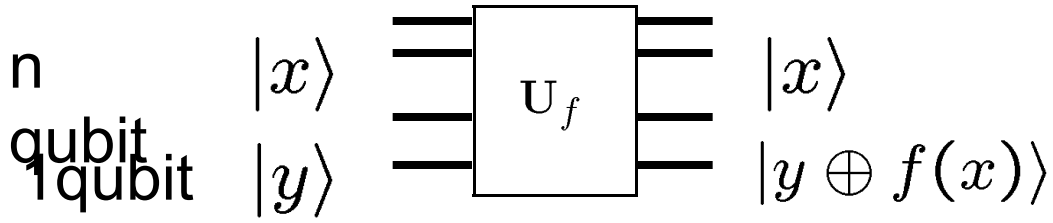
$$Pr(fail) = \frac{1}{2} - \frac{1}{2^n} < \frac{1}{2}$$

Bounded error classical q. complexity: $O(1)$

Quantum Deutsch-Jozsa

Given: A quantum gate on n+1 qubits strings which calculates the promised f

$$f : x \in \{0,1\}^n \rightarrow \{0,1\}$$



$$y, f(x) \in \{0,1\}$$

$$x \in \{0,1\}^n$$

$$|x\rangle \otimes |y\rangle \xrightarrow{\mathbf{U}_f} |x\rangle \otimes |y \oplus f(x)\rangle$$

Trick 1: Phase Kickback



$$|x\rangle \otimes |y\rangle \overset{\mathbf{U}_f}{\rightarrow} |x\rangle \otimes |y \oplus f(x)\rangle$$

Input a superposition over second register:

$$|x\rangle \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \overset{\mathbf{U}_f}{\rightarrow} |x\rangle \otimes \frac{1}{\sqrt{2}}(|f(x)\rangle - |\bar{f}(x)\rangle)$$

$$f(x) = 0 \quad |x\rangle \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

$$f(x) = 1 \quad -|x\rangle \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

Function is computed into phase:

$$|x\rangle \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \overset{\mathbf{U}_f}{\rightarrow} (-1)^{f(x)}|x\rangle \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

Trick 2: Hadamarding Qubits





$$|x\rangle \;\boxed{H}\; \frac{1}{\sqrt{2}}(|0\rangle + (-1)^x|1\rangle)$$

$$|x_1\rangle \;\boxed{H}$$

$$|x_2\rangle \;\boxed{H}$$

$$|x_3\rangle \;\boxed{H}$$

$$\vdots$$

$$|x_n\rangle \;\boxed{H}$$

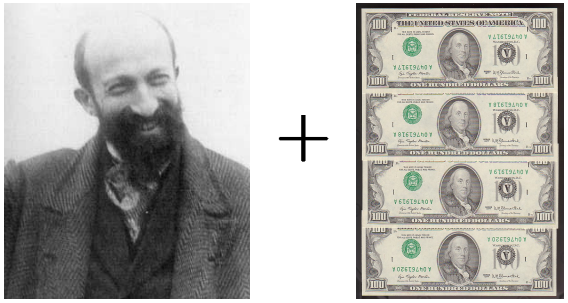
$$|x\rangle = |x_1, x_2, \ldots, x_n\rangle$$

$$\frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} (-1)^{y \cdot x} |y\rangle$$

$$y \cdot x = y_1 x_1 \oplus y_2 x_2 \oplus \cdots \oplus y_n x_n$$

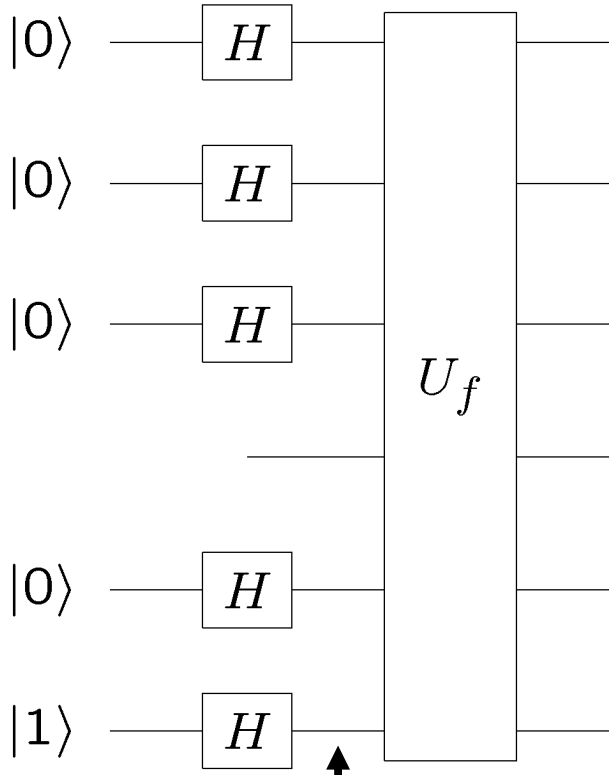
Note: $H^{\otimes n}|0\rangle = \frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} |y\rangle$ and $H^{\otimes n} \frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} |y\rangle = |0\rangle$

Tricks 1 and 2 Together



$$\frac{1}{\sqrt{2^n}}\sum_{y=0}^{2^n-1}(-1)^{f(y)}|y\rangle\frac{1}{\sqrt{2}}(|0\rangle-|1\rangle)$$

n qubits



$$H^{\otimes n}|0\rangle\otimes H|1\rangle=\frac{1}{\sqrt{2^n}}\sum_{y=0}^{2^n-1}|y\rangle\frac{1}{\sqrt{2}}(|0\rangle-|1\rangle)$$

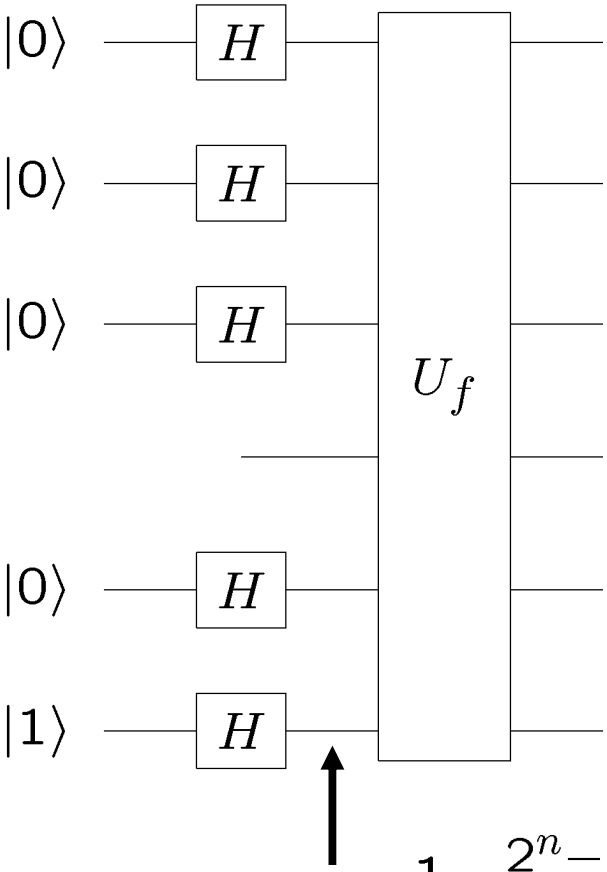
Tricks 1 and 2 Together



+



n qubits



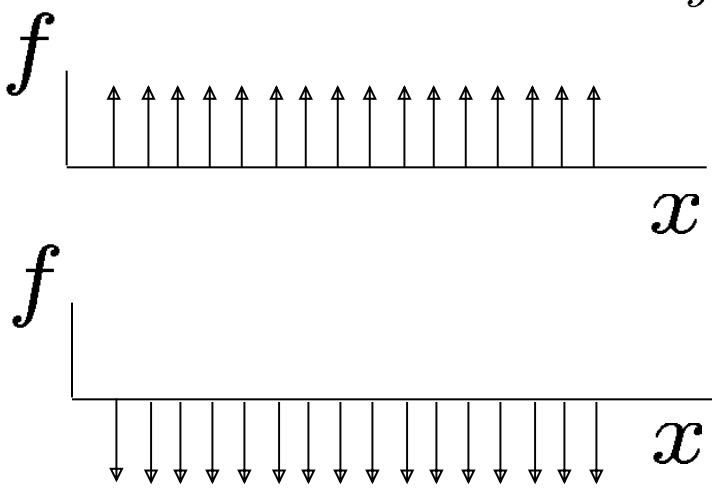
$$\frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} (-1)^{f(y)} |y\rangle$$

$$\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

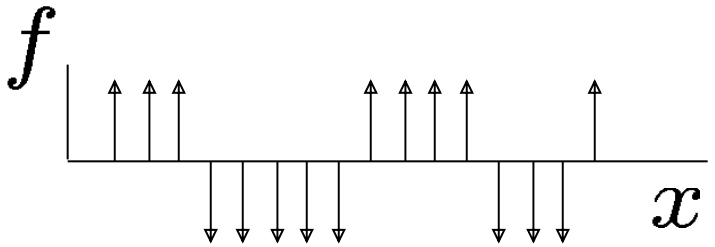
$$H^{\otimes n}|0\rangle \otimes H|1\rangle = \frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} |y\rangle \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

Function in the Phase

$$\frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} (-1)^{f(y)} |y\rangle$$



constant



balanced

$$\frac{(-1)^{f(0)}}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} |y\rangle$$

Function in the Phase

When the function is constant:

$$H^{\otimes n}\left[\frac{(-1)^{f(0)}}{\sqrt{2^n}}\sum_{y=0}^{2^n-1}|y\rangle\right] = (-1)^{f(0)}|0\rangle$$

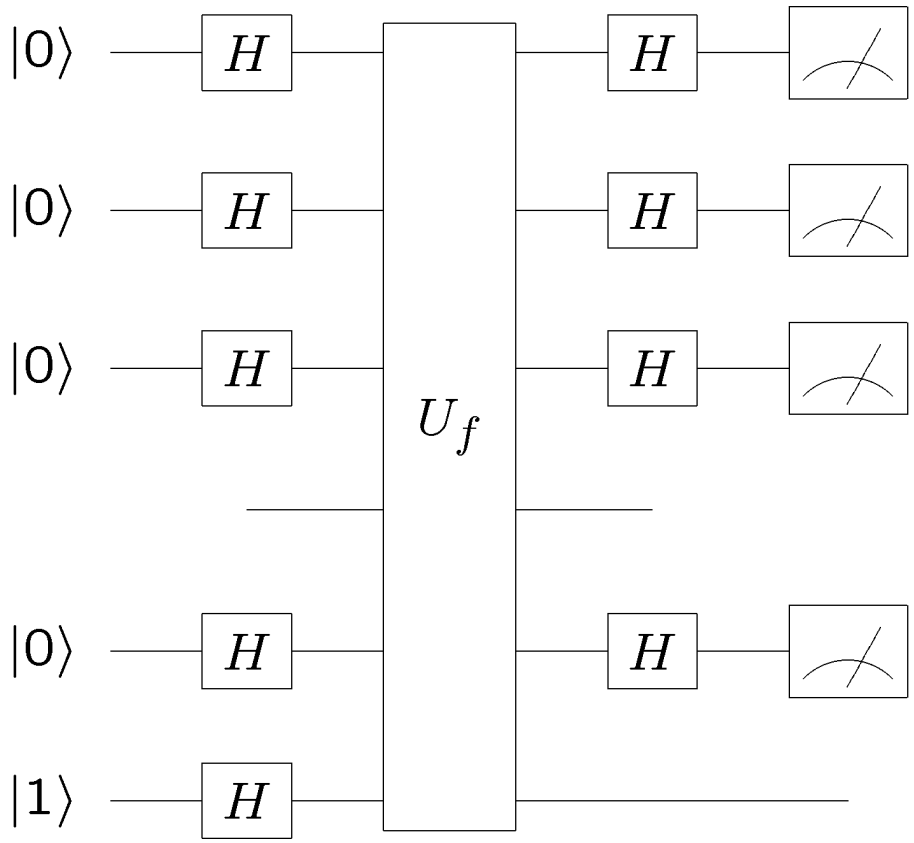
When the function is balanced:

$$H^{\otimes n}\left[\frac{1}{\sqrt{2^n}}\sum_{y=0}^{2^n-1}(-1)^{f(y)}|y\rangle\right] = \frac{1}{2^n}\sum_{x,y=0}^{2^n-1}(-1)^{f(y)+y\cdot x}|x\rangle$$

amplitude in zero state

$$\langle 0|H^{\otimes n}\left[\frac{1}{\sqrt{2^n}}\sum_{y=0}^{2^n-1}(-1)^{f(y)}|y\rangle\right] = \frac{1}{2^n}\sum_{y=0}^{2^n-1}(-1)^{f(y)} = 0$$

Quantum Deutsch-Jozsa



n qubits

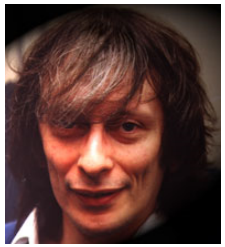
r

If function is constant, r is always 0.

If function is balanced, r is never 0.

Distinguish constant from balanced using one quantum query

Deutsch-Jozsa



David Deutsch



Richard Jozsa

1992: Deutsch-Jozsa Algorithm

Exact classical q. complexity: $2^{n-1}+1$

Bounded error classical q. complexity: $O(1)$

Exact quantum q. complexity: 1

Bernstein-Vazirani Problem

Given: A function with n bit strings as input and one bit as output

$$f : x \in \{0,1\}^n \rightarrow \{0,1\}$$

Promise: The function is of the form

$$f(x) = (a \cdot x) \oplus b \qquad a \in \{0,1\}^n \quad b \in \{0,1\}$$

$$y \cdot x = y_1 x_1 \oplus y_2 x_2 \oplus \cdots \oplus y_n x_n$$

Problem: Find the n bit string a

Classical Bernstein-Vazirani

Given: A function with n bit strings as input and one bit as output

$$f : x \in \{0,1\}^n \rightarrow \{0,1\}$$

Promise: The function is of the form

$$f(x) = (a \cdot x) \oplus b \qquad a \in \{0,1\}^n \quad b \in \{0,1\}$$

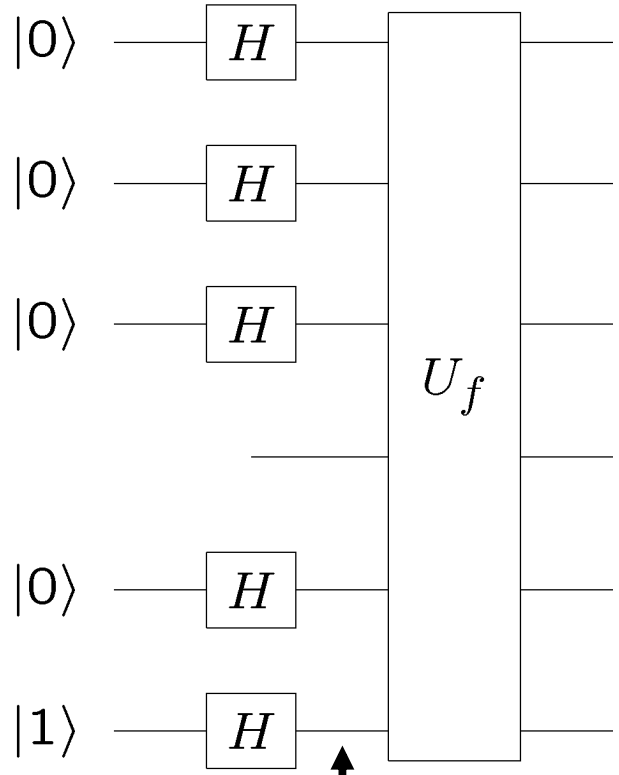
$$y \cdot x = y_1 x_1 \oplus y_2 x_2 \oplus \cdots \oplus y_n x_n$$

Problem: Find the n bit string a

Notice that the querying f yields a single bit of information. But we need n bits of information to describe a.

Bounded error classical q. complexity: $\Omega(n)$

Quantum Bernstein-Vazirani



n qubits

$$\frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} (-1)^{f(y)} |y\rangle$$

$$= \frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} (-1)^{a \cdot y + b} |y\rangle$$

$$\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

$$H^{\otimes n}|0\rangle \otimes H|1\rangle = \frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} |y\rangle \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

Hadamard It!

$$H^{\otimes n}\left[\frac{1}{\sqrt{2^n}}\sum_{y=0}^{2^n-1}(-1)^{a\cdot y+b}|y\rangle\right]=\frac{1}{2^n}\sum_{x,y=0}^{2^n-1}(-1)^{a\cdot y+b}(-1)^{x\cdot y}|x\rangle$$

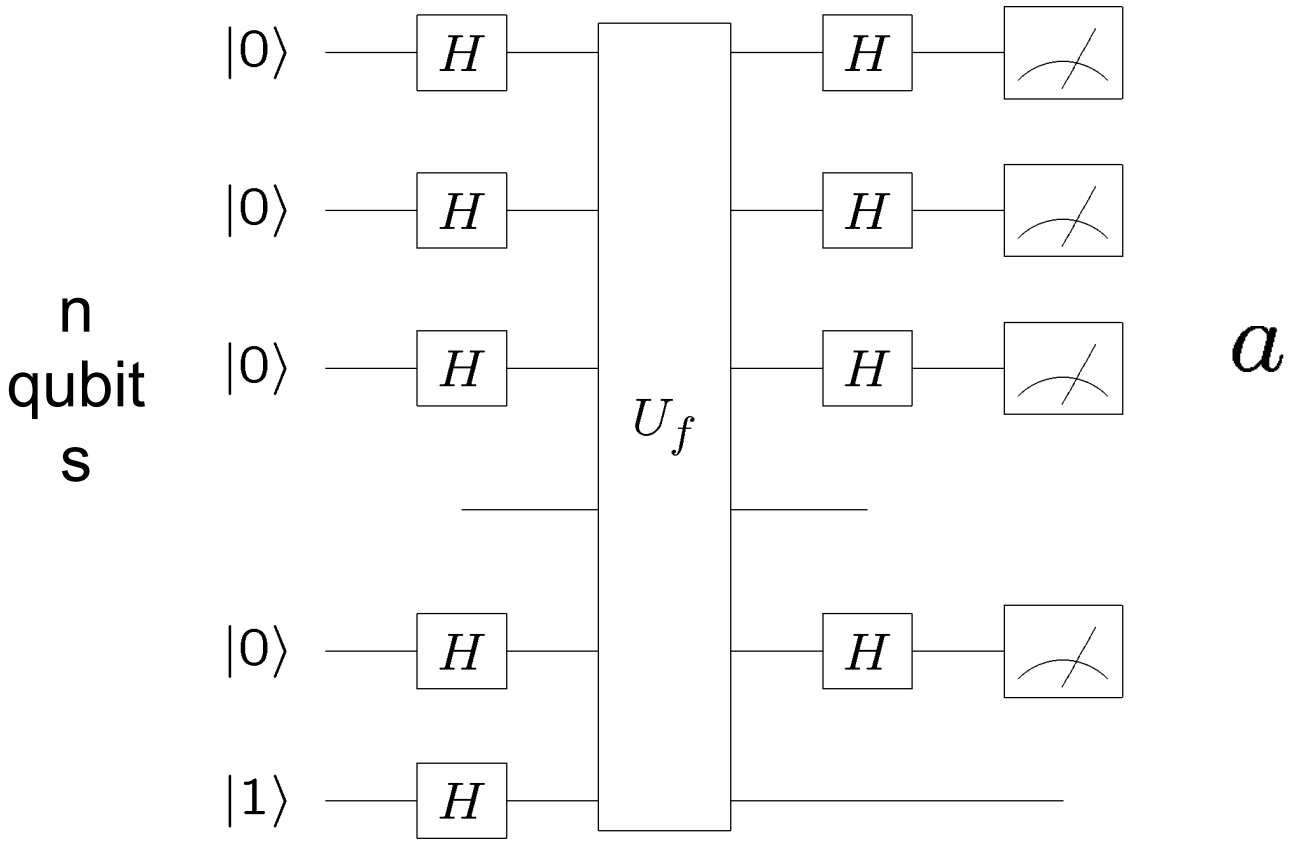
$$=\frac{(-1)^b}{2^n}\sum_{x=0}^{2^n-1}\left(\sum_{y=0}^{2^n-1}(-1)^{a\cdot y+x\cdot y}\right)|x\rangle$$

$$\sum_{y=0}^{2^n-1}(-1)^{a\cdot y+x\cdot y}=\sum_{y_1,\ldots,y_n=0}^{n}(-1)^{a_1\oplus y_1+\cdots+a_n\oplus y_n+x_1\oplus y_1+x_n\oplus y_n}$$

$$=2^n\delta_{a_1,x_1}\delta_{a_2,x_2}\cdots\delta_{a_n,x_n}$$

$$H^{\otimes n}\left[\frac{1}{\sqrt{2^n}}\sum_{y=0}^{2^n-1}(-1)^{a\cdot y+b}|y\rangle\right]=(-1)^b|a\rangle$$

Quantum Bernstein-Vazirani



We can determine a using only a single quantum query!

Bernstein-Vazirani



Umesh Vazirani



Ethan Bernstein

1993: Bernstein-Vazirani Algorithm (non-recursive)

Exact classical q. complexity: n

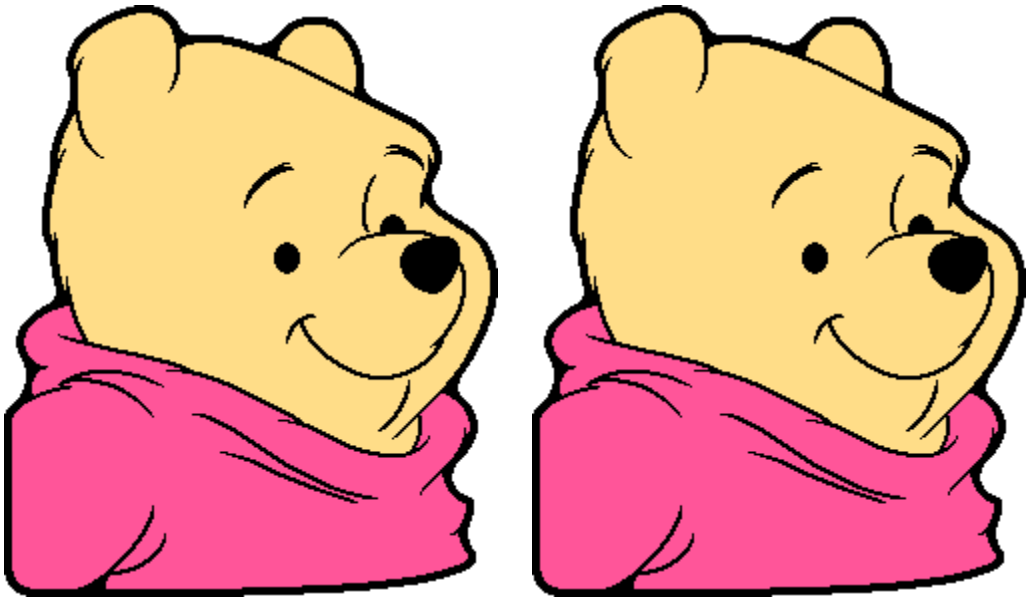
Bounded error classical q. complexity: $\Omega(n)$

Exact quantum q. complexity: 1

Pooh-Pooh?

People like to pooh-pooh these early problems because they do not solve problems which are “natural”

This is silly. These results show that treating a device as classical or as quantum show amazing differences and should only be “pooh-pooh”ed by people who have invented their own new quantum algorithms.



Simon's Problem

(is that nobody does what Simon says)

NORTON SIMON MUSEUM

Given: A function with n bit strings as input and one bit as output

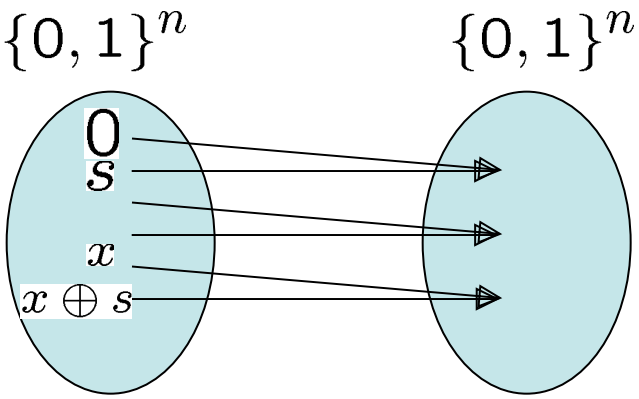
$$f : x \in \{0,1\}^n \rightarrow \{0,1\}^n$$

Promise: The function is guaranteed to satisfy

$$f(x) = f(y) \Rightarrow y = x \oplus s$$

$$x \oplus s = (x_1 \oplus s_1, x_2 \oplus s_2, \ldots, x_n \oplus s_n)$$

Problem: Find the n bit string s



Classical Simon's Problem

Promise: The function is guaranteed to satisfy

$$f(x) = f(y) \Rightarrow y = x \oplus s$$

$$x \oplus s = (x_1 \oplus s_1, x_2 \oplus s_2, \ldots, x_n \oplus s_n)$$

Suppose we start querying the function and build up a list of the pairs $(x_\alpha, f(x_\alpha))$

If we find $x_\alpha \neq x_\beta$ such that $f(x_\alpha) = f(x_\beta)$ then we solve the problem:

$$f(x_\alpha) = f(x_\beta) \Rightarrow x_\alpha = s \oplus x_\beta$$

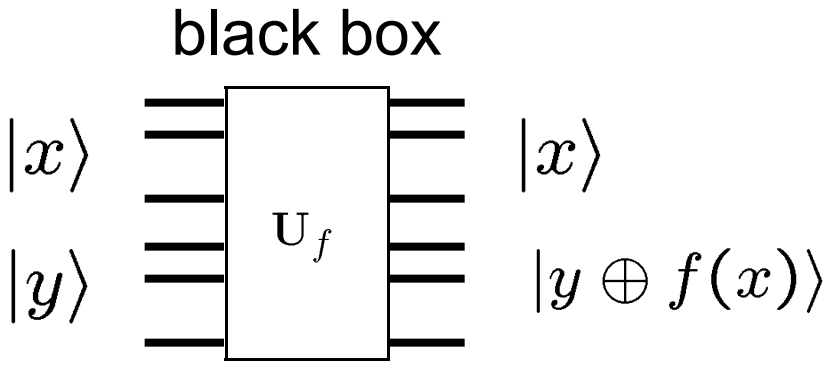
$$s = x_\alpha \oplus x_\beta$$

But suppose we start querying the function m times….

Probability of getting a matching pair $\approx \dfrac{\binom{m}{2}}{2^n}$

Bounded error query complexity: $m = O\left(2^{\frac{n}{2}}\right)$

Quantum Simon's Problem



Unlike previous problems, we can't use the phase kickback trick because there is no structure in the function.

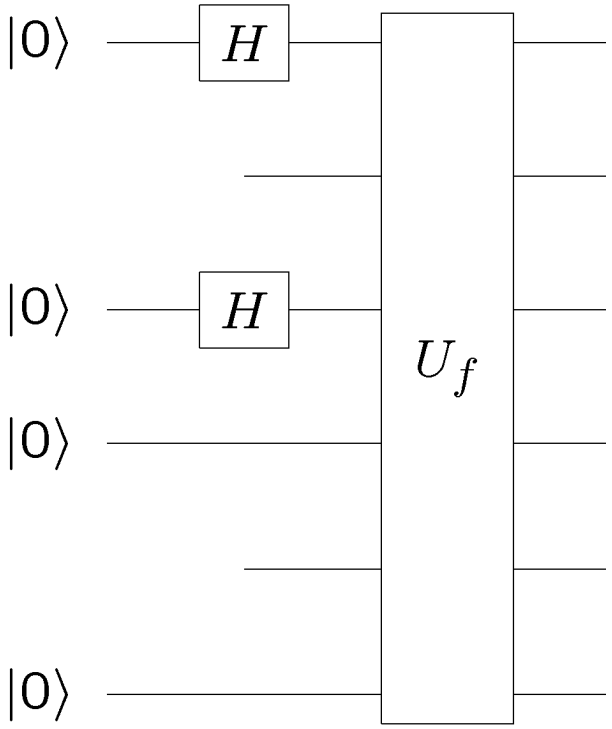
Charge ahead:



Quantum Simon's Problem

n qubits

n qubits



$$\frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle \otimes |0\rangle$$

$$\frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle \otimes |f(x)\rangle$$

Quantum Simon's Problem

Measure the second register

$$\frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle \otimes |f(x)\rangle$$

Using the promise on the function

$$f(x) = f(y) \Rightarrow y = x \oplus s$$

This implies that we have the state

$$\frac{1}{\sqrt{2}}(|x\rangle + |x \oplus s\rangle)$$

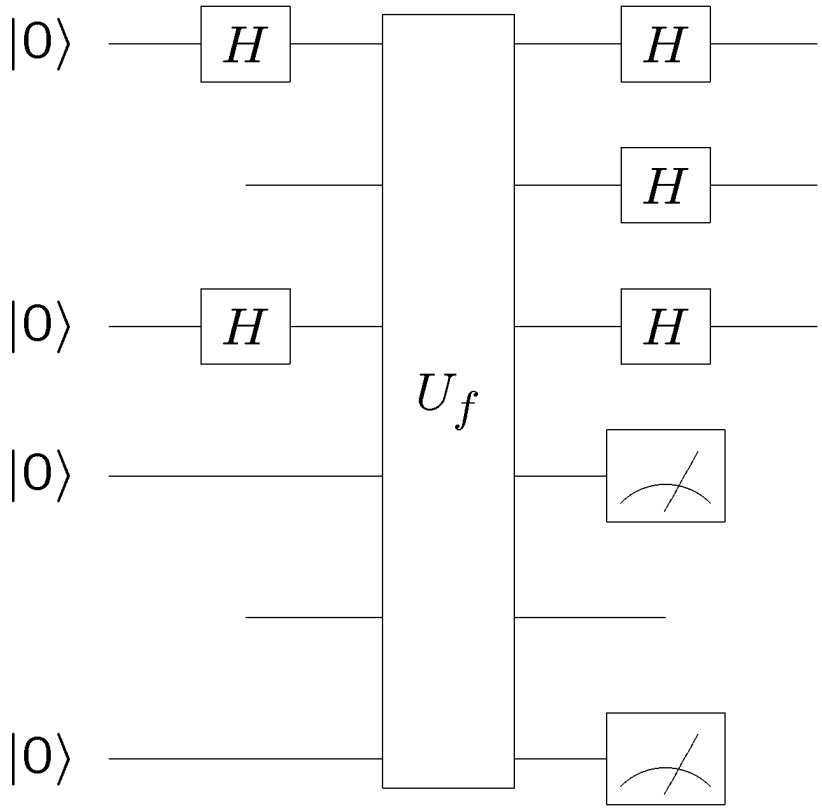
For random uniformly distributed x

Measuring this state at this time does us no good….

Quantum Simon's Problem

n qubits

n qubits



$$H^{\otimes n}\frac{1}{\sqrt{2}}(|x\rangle+|x\oplus s\rangle) = \frac{1}{\sqrt{2^{n+1}}}\sum_{y=0}^{2^n-1}((-1)^{y\cdot x}+(-1)^{y\cdot(x\oplus s)})|y\rangle$$

Quantum Simon's Problem

$$H^{\otimes n}\frac{1}{\sqrt{2}}(|x\rangle+|x\oplus s\rangle) = \frac{1}{\sqrt{2^{n+1}}}\sum_{y=0}^{2^n-1}((-1)^{y\cdot x}+(-1)^{y\cdot(x\oplus s)})|y\rangle$$

$$y\cdot x = y_1x_1 \oplus y_2x_2 \oplus \cdots \oplus y_nx_n$$

$$y\cdot(x\oplus s) = y_1(x_1\oplus s_1)\oplus y_2(x_2\oplus s_2)\oplus\cdots\oplus y_n(x_n\oplus s_n) = (y\cdot x)\oplus(y\oplus s)$$

$$= \frac{1}{\sqrt{2^{n+1}}}\sum_{y=0}^{2^n-1}(-1)^{y\cdot x}(1+(-1)^{y\cdot s})|y\rangle$$

Measuring this state, we obtain uniformly distributed random values of y such that

$$y\cdot s = 0$$

If $y \neq 0$ we have eliminated the possible values of s by half

Quantum Simon's Problem

Think about the bit strings s as vectors in $\mathbb{Z}_2^n$

Multiple runs of the quantum algorithm yield equations

$$\begin{aligned} y_0 \cdot s &= 0 \\ y_1 \cdot s &= 0 \\ &\vdots \\ y_k \cdot s &= 0 \end{aligned}$$

random uniform y_i

If we obtain n linearly independent equations of this form, we win (Guassian elimination)

Suppose we have k linearly independent y_i 's. What is the probability that y_{k+1} is linearly independent of previous y_i 's?

$$Pr = \frac{2^n - 2^k}{2^n} = 1 - 2^{k-n}$$

Quantum Simon's Problem

$$Pr = \frac{2^n - 2^k}{2^n} = 1 - 2^{k-n}$$

What is the probability that our n equations are linearly independent?

$$Pr(succ) = \left(1 - \frac{1}{2^n}\right)\left(1 - \frac{1}{2^{n-1}}\right)\cdots\left(1 - \frac{1}{2}\right)$$

$$= \prod_{k=1}^{n}\left(1 - \frac{1}{2^k}\right) > \prod_{k=1}^{\infty}\left(1 - \frac{1}{2^k}\right) \approx 0.28879$$

With constant probability we obtain linearly independence and hence solve Simon's problem.

Quantum Simon's Problem



Dan Simon

1994: Simon's Algorithm

Bounded error classical q. complexity: $\Omega(2^{n/2})$

Bounded error quantum q. complexity: $O(n)$

It is also possible to make Simon's algorithm an exactly quantum algorithm with the same query complexity.

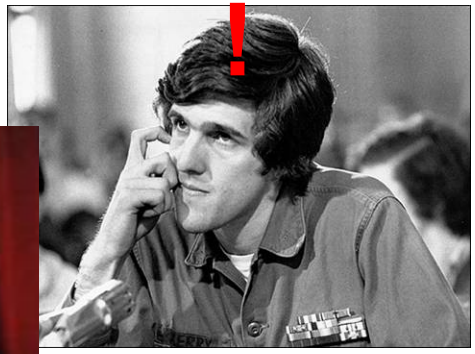
Further note that the actually running time for Simon's quantum algorithm has a running time $O(n^{2.376})$ using the best known algorithms for solving the system of linear equations

A Game

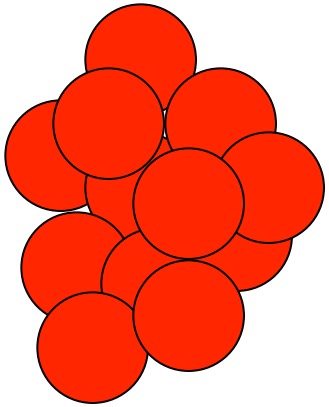
winner !

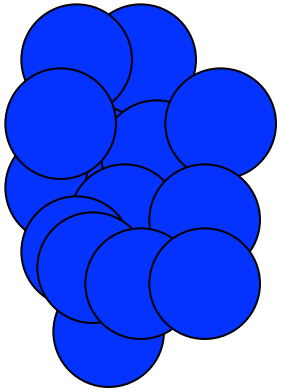


winner !









winner is last to place disk

The Physicist's Game

Mmmm…Symmetry



What are physicists good at?

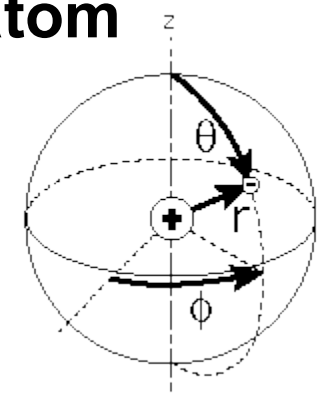
symmetry

symmetry

Why are physicists good at this?

1. **Nature is often symmetric!**
2. **Symmetric problems are solvable problems, especially in quantum theory!**

Example: Hydrogen atom



$$\left[-\frac{\hbar^2}{2\mu}\nabla^2 - \frac{e^2}{r}\right]\psi(x,y,z) = E\psi(x,y,z)$$

spherical symmetry

$$-\frac{\hbar^2}{2\mu r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) + \left[\frac{\hbar^2 l(l+1)}{2\mu r^2} - \frac{e}{r} - E\right]R = 0$$

Mmmm…Symmetry



Simon's algorithm is an example of a problem for which the promise on the function is a symmetry

$$f(x) = f(y) \Rightarrow y = x \oplus s$$

Random quantum states

$$\frac{1}{\sqrt{2}}(|x\rangle + |x \oplus s\rangle)$$

which is the density matrix

$$\rho = \frac{1}{2^n} \sum_{x=0}^{2^n} \frac{1}{\sqrt{2}}(|x\rangle + |x \oplus s\rangle)\frac{1}{\sqrt{2}}(\langle x| + \langle x \oplus s|)$$

Define the symmetry operators $U_y|x\rangle = |y \oplus x\rangle$

Then the density matrix is invariant under this symmet$U_y \rho U_y^\dagger = \rho$

Shor



Peter Shor 1994

188198812920607963838697239461650439807163563379417382700763356422988859715234665485319060606504743045317388011303396716199692321205734031879550656996221305168759307650257059

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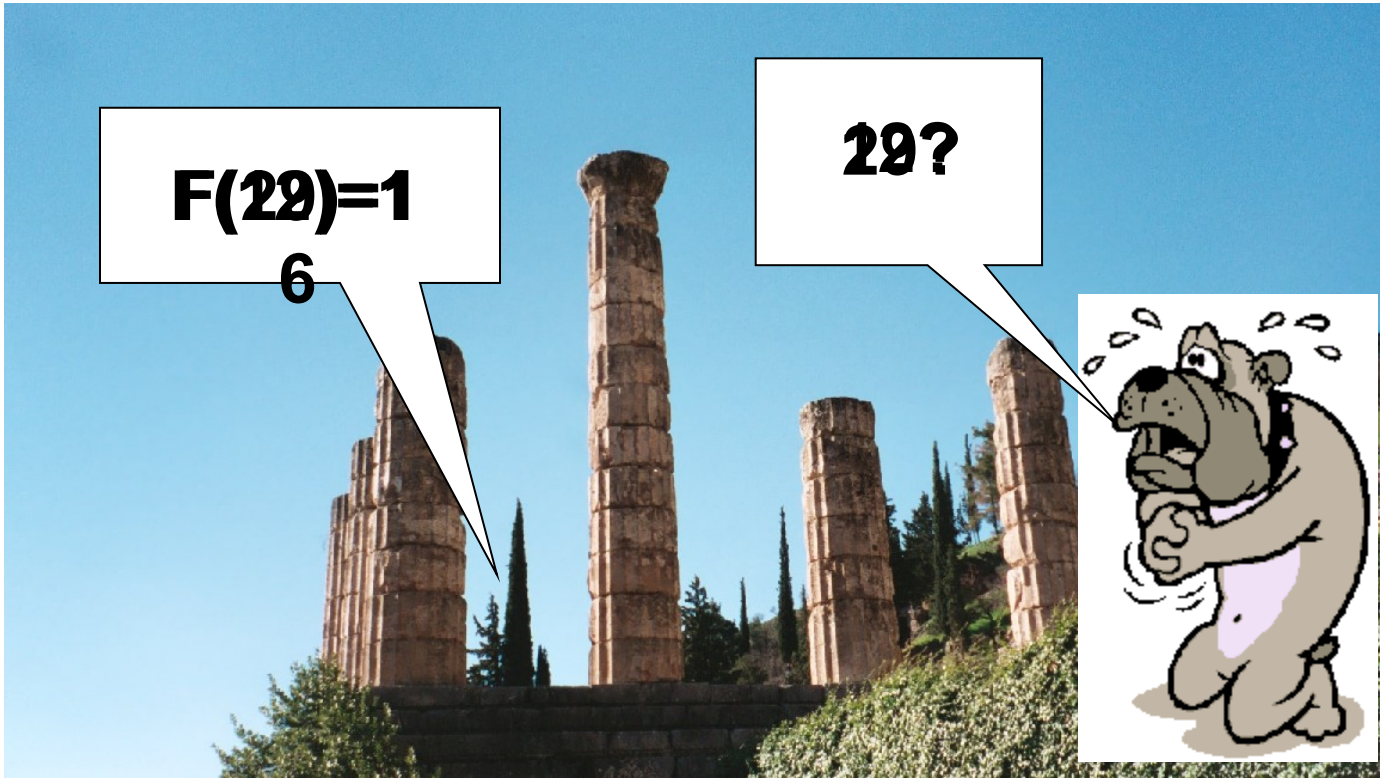
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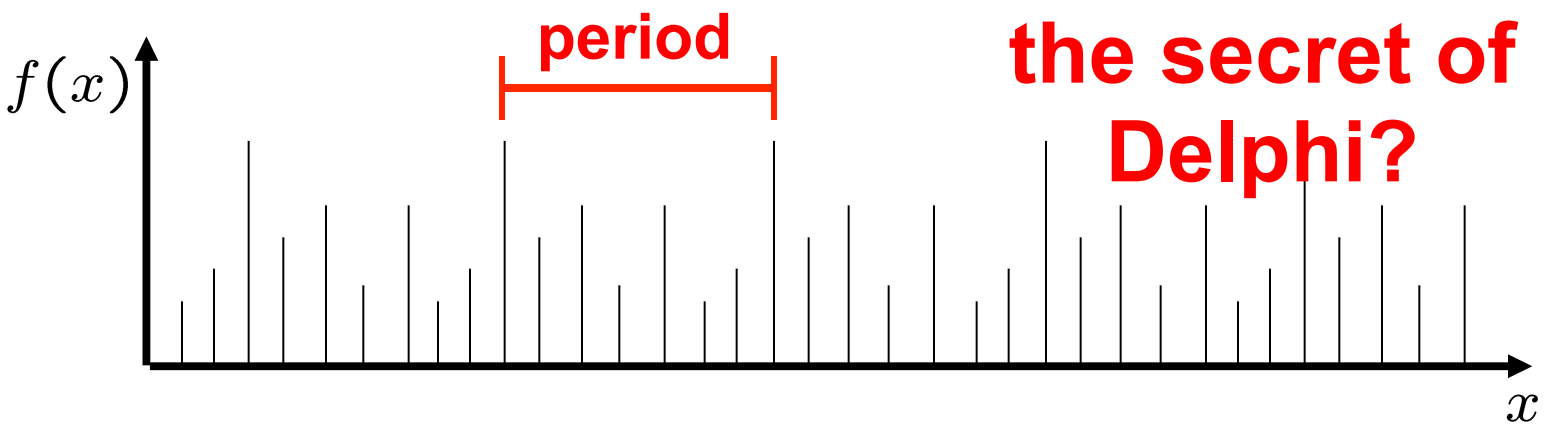
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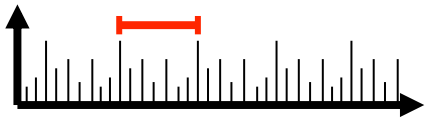
Shor's algorithm for factoring is a combination of the insight
that quantum computers are good at solving symmetric problems combined with the number theoretic observation that factoring can be solved by period finding.

A Guessing Game

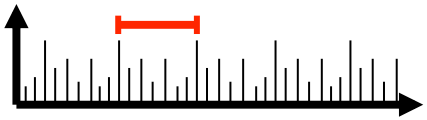




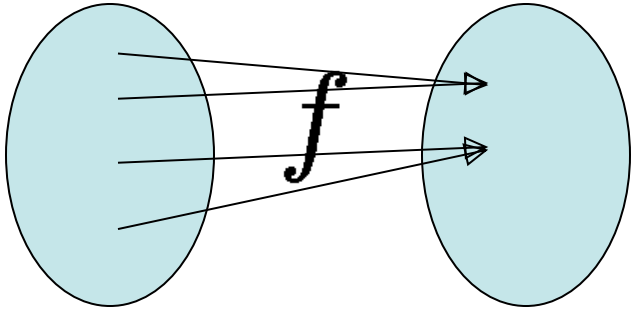
the secret of Delphi?



Period Finding



$x \in \{0,1\}^n \qquad y \in \{0,1\}^n$

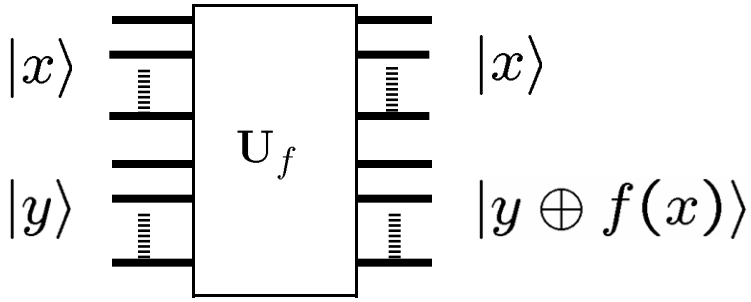


$$f(x) = f(x+l)$$

Problem: find l in as few queries as possible

Period Finding Problem

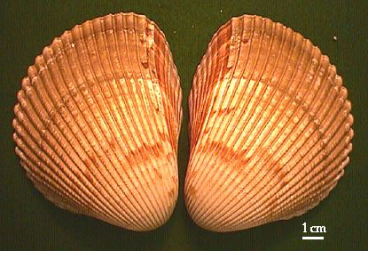
quantum oracle



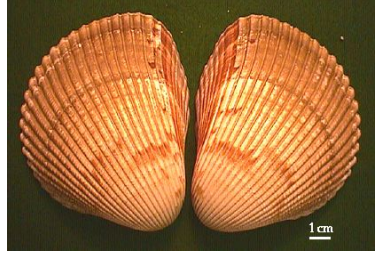
$x, y, f(x) \in \{0,1\}^n$

….in as few uses of the quantum oracle as possible

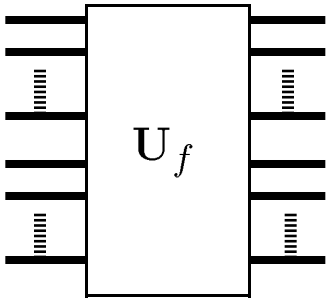
a symmetric problem!



Explicit Symmetry



$$\frac{1}{\sqrt{2^n}}\sum_{x=0}^{2^n-1}|x\rangle \qquad |0\rangle$$



$$\frac{1}{\sqrt{2^n}}\sum_{x=0}^{2^n-1}|x\rangle\otimes|f(x)\rangle$$

measure second register, randomly obtain $f(x_0)$

first register:

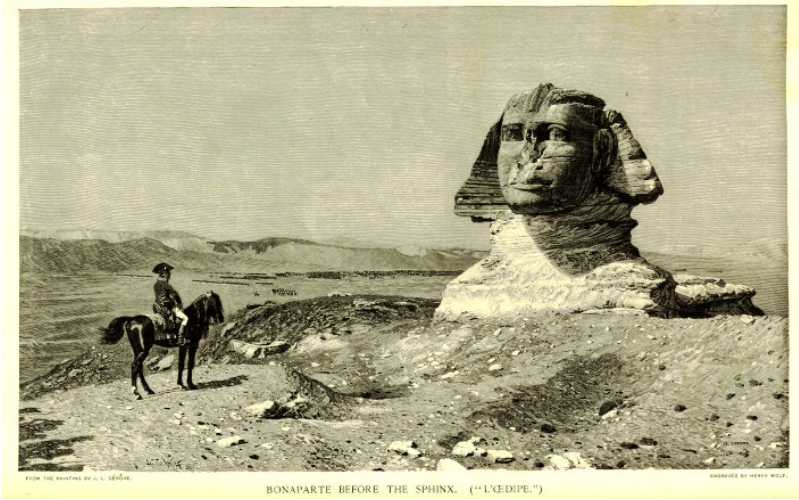
$$|\psi(x_0)\rangle=\frac{1}{\sqrt{2^n/l}}\sum_{j=0}^{2^n/l-1}|x_0+jl\rangle \Longrightarrow \boldsymbol{\rho}=\frac{1}{l}\sum_{x_0=0}^{l-1}|\psi(x_0)\rangle\langle\psi(x_0)|$$

$$\frac{1}{\sqrt{2^n/l}}\sum_{j=0}^{2^n/l-1}\mathbf{T}^k|x_0+jl\rangle=\frac{1}{\sqrt{2^n/l}}\sum_{j=0}^{2^n/l-1}|x_0+k+jl\rangle=\frac{1}{\sqrt{2^n/l}}\sum_{j=0}^{2^n/l-1}|x_0'+jl\rangle$$

$$\mathbf{T}^k\boldsymbol{\rho}\mathbf{T}^{-k}=\boldsymbol{\rho}$$

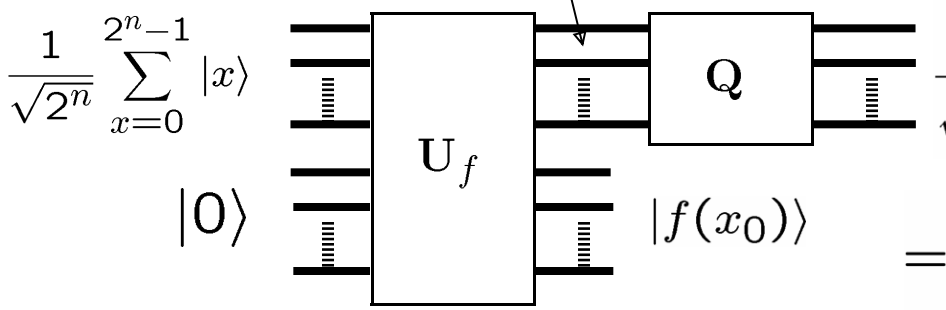
translational invariance $\Longrightarrow$ work in momentum basis

Fourier to the Rescue



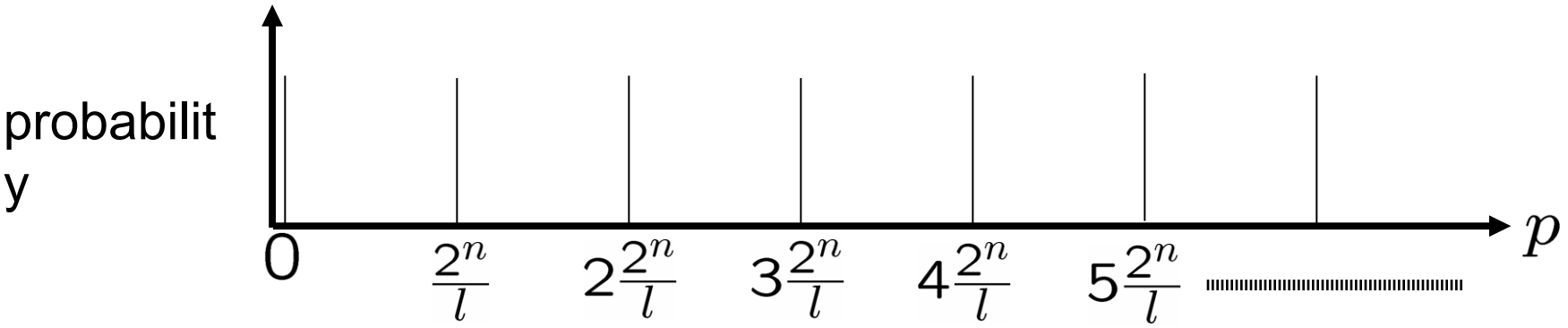
$$\mathbf{Q} = \frac{1}{\sqrt{2^n}} \sum_{x,p=0}^{2^n-1} \exp\left(\frac{2\pi i p x}{2^n}\right) |p\rangle\langle x|$$

$$|\psi(x_0)\rangle = \frac{1}{\sqrt{2^n/l}} \sum_{j=0}^{2^n/l-1} |x_0 + jl\rangle$$



$$\frac{1}{\sqrt{2^{2n}/l}} \sum_{p=0}^{2^n-1} \sum_{j=0}^{2^n/l-1} \exp\left(\frac{2\pi i p(x_0 + jl)}{2^n}\right) |p\rangle$$

$$= \frac{1}{\sqrt{l}} \sum_{p=0|pl=0 \bmod 2^n}^{2^n-1} e^{\frac{2\pi i p x_0}{2^n}} |p\rangle$$



The Factoring Firestorm



Peter Shor 1994

188198812920607963383697239461650439807163563379417382700763356422988859715234665485319060606504743045317388011303396716199623212057340318795506569622130516875930765025705 9

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3980750864240649373971255005503864911990643623425267084063851895759463889572617685833172727214610743530253622307197304822463291469530209711645985217113052071125636359039752 7

×

Best classical algorithm takes time $O(\exp(n^{1/3}))$

Shor's quantum algorithm takes time $O(n^2 \log n)$

An efficient algorithm for factoring **breaks** the RSA public key cryptosystem

Shor's Algorithm

Breaks RSA Cryptography!



"Only time will tell if and when the problems of building a quantum computer can be overcome….As information becomes the world's most valuable commodity, the economic, political and military fate of nations will depend on the strength of ciphers. Consequently, the development of a fully operational quantum computer would imperil our personal privacy, destroy electronic commerce and demolish the concept of national security. A quantum computer would jeopardize the stability of the world. Whichever country gets there first will have the ability to monitor the communications of its citizens, read the minds of its commercial rivals and eavesdrop on the plans of its enemies. Although it is still in its infancy, quantum computing presents a potential threat to the individual, to international business and to global security. "

-Simon Singh

This Space for Sale



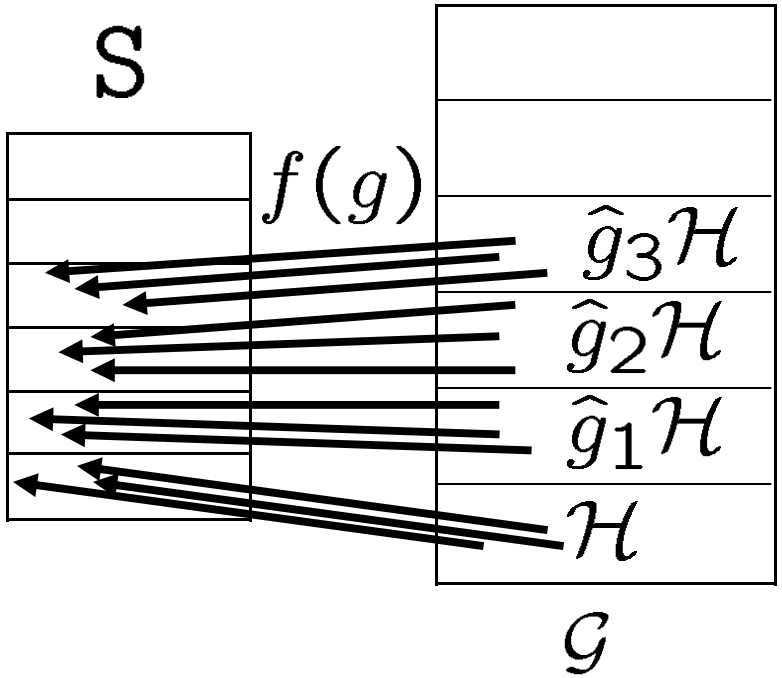
The Hidden Subgroup Problem

Finite group: $\mathcal{G}$ $\quad g_1 g_2 = g_3,\ g_1 g_1^{-1} = e,$ etc.

Given: $f(g)$ $\quad g \in \mathcal{G} \rightarrow \mathrm{S}$

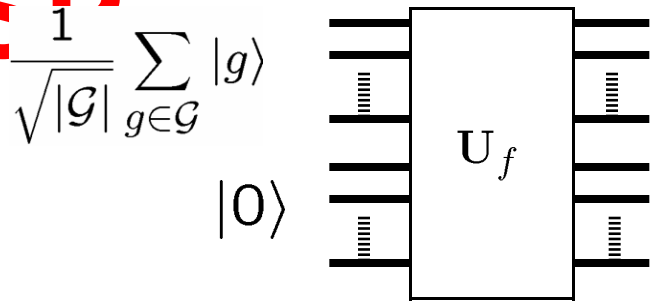
which is constant and distinct on left cosets of some subgroup $\mathcal{H}$

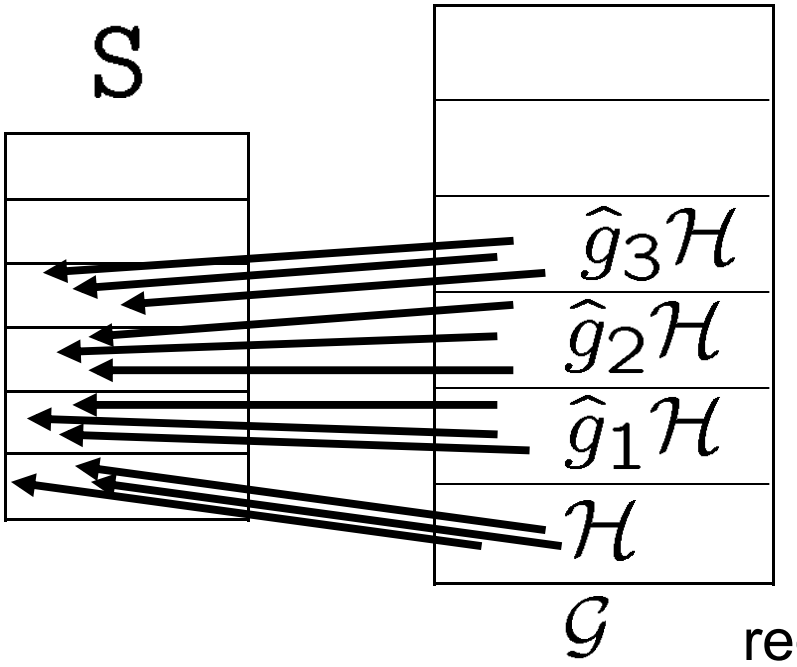
Find: generators for $\mathcal{H}$



$$\hat{g}_3\mathcal{H} = \{\hat{g}_3, \hat{g}_3 h_1, \hat{g}_3 h_2, \ldots\}$$
$$\hat{g}_2\mathcal{H} = \{\hat{g}_2, \hat{g}_2 h_1, \hat{g}_2 h_2, \ldots\}$$
$$\hat{g}_1\mathcal{H} = \{\hat{g}_1, \hat{g}_1 h_1, \hat{g}_1 h_2, \ldots\}$$
$$\mathcal{H} = \{e, h_1, h_2, \ldots\}$$

Symmetry and the HSP





function register $f(\hat{g}_i)$

outcome $|\hat{g}_i\mathcal{H}\rangle = \frac{1}{\sqrt{|\mathcal{H}|}}\sum_{h\in\mathcal{H}}|\hat{g}_i h\rangle$

$$\rho_{\mathcal{H}} = \frac{|\mathcal{H}|}{|\mathcal{G}|}\sum_i |g_i\mathcal{H}\rangle\langle g_i\mathcal{H}|$$

$$\mathbf{D}_L(g)\rho\mathbf{D}_L(g^{-1}) = \rho$$

regular representation: $\mathbf{D}_L(g)|g'\rangle = |gg'\rangle$

SYMMETRY!

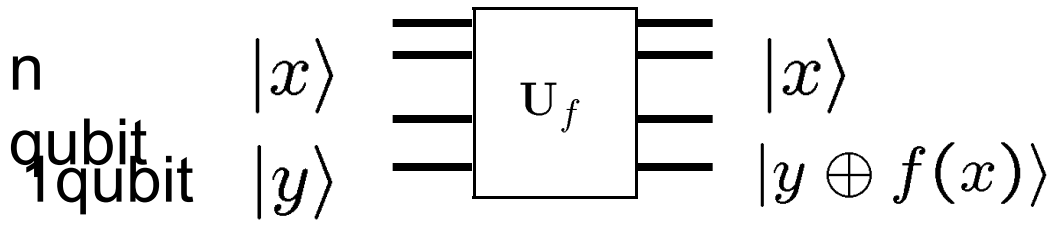
We Care About HSP Because…

Group $\mathcal{G}$.	classical problem
Abelian	
$\mathbb{Z}_n$	period finding order finding factoring
$\mathbb{Z}_n \times \mathbb{Z}_n$	discrete logarithm
Abelian groups	*efficiently solvable*
	efficiently solvable?
$\mathcal{D}_N$ dihedral group	certain shortest vector in a lattice problems
$\mathcal{S}_N$ symmetric group	graph isomoprhism
non-Abelian	

Algorithms of the Grover Type

Grover's Problem

Suppose we have a black box



with the property

$$f(x_0) = 1$$

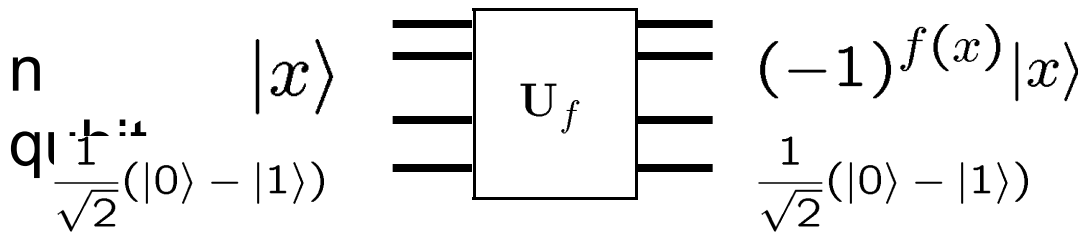
$$\forall x \neq x_0, f(x) = 0$$

Problem: find x_0 with as few queries as possible.



Grover's Algorithm

Use the black box in a particular way



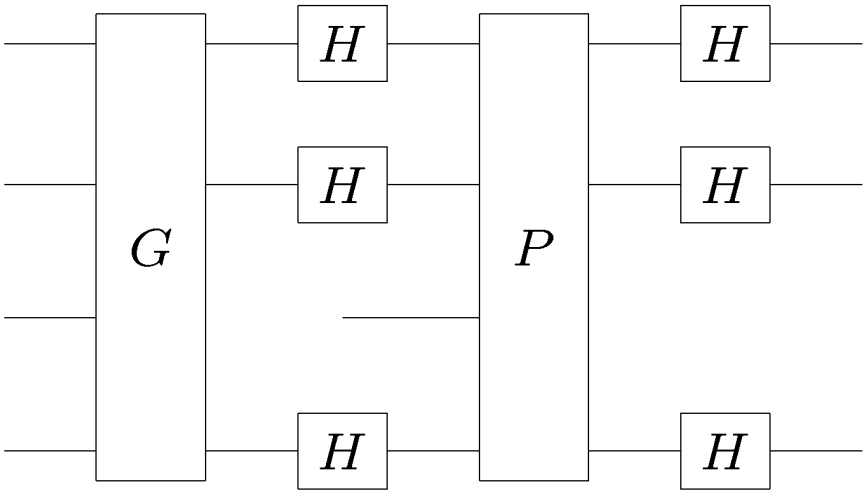
Grover oracle:

$$G = \sum_{x=0}^{2^n} (-1)^{f(x)} |x\rangle\langle x|$$

How to use Grover oracle to find $f(x_0) = 1$?

The Grover Iterate

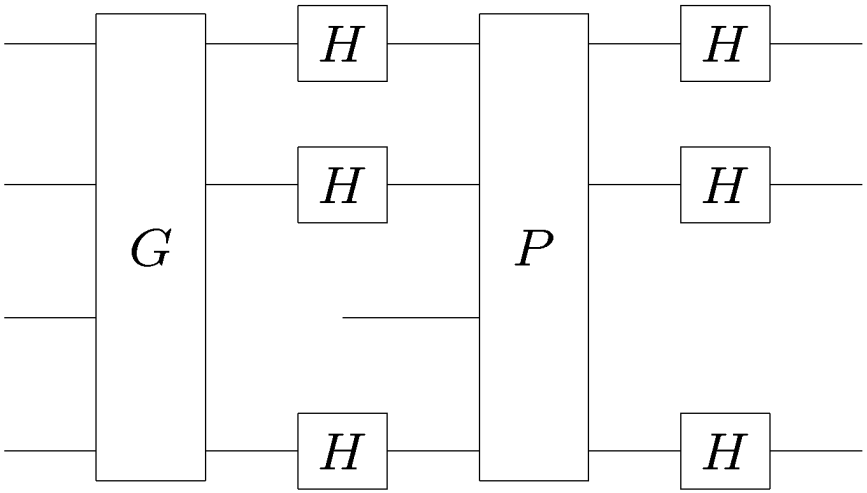
n
qubits



$$G = \sum_{x=0}^{2^n} (-1)^{f(x)} |x\rangle\langle x| \qquad P = \sum_{x=0}^{2^n} -(-1)^{\delta_{x,0}} |x\rangle\langle x|$$

The Grover Iterate

n
qubits



$$P = \sum_{x=0}^{2^n} -(-1)^{\delta_{x,0}}|x\rangle\langle x| = 2|0\rangle\langle 0| - I$$

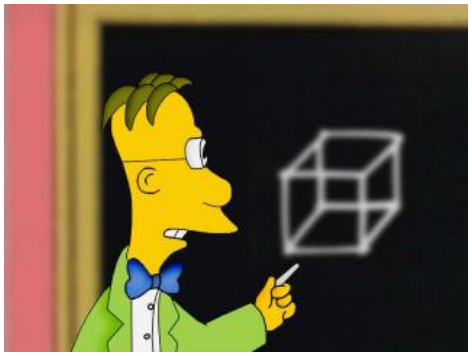
$$H^{\otimes n} P H^{\otimes n} = H^{\otimes n}|0\rangle\langle 0|H^{\otimes n} - I = 2|\psi\rangle\langle\psi| - I$$

$$|\psi\rangle = \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle$$

Grover's iterate

$$R = (2|\psi\rangle\langle\psi| - I)G$$

The Grover Iterate in 2D



Two orthonormal vectors:

$$|\alpha\rangle = \frac{1}{\sqrt{2^n - 1}} \sum_{x=0|f(x)=0}^{2^n-1} |x\rangle \qquad |x_0\rangle$$

Express the equal superposition in terms of these:

$$|\psi\rangle = \sqrt{\frac{2^n - 1}{2^n}}|\alpha\rangle + \frac{1}{\sqrt{2^n}}|x_0\rangle$$

The Grover iterate will preserve this two dimensional subspace

$$G = \sum_{x=0}^{2^n} (-1)^{f(x)}|x\rangle\langle x| \qquad \begin{aligned} G|\alpha\rangle &= |\alpha\rangle \\ G|x_0\rangle &= -|x_0\rangle \end{aligned}$$

$$2|\psi\rangle\langle\psi| - I = 2\left[\left(\sqrt{\frac{2^n - 1}{2^n}}|\alpha\rangle + \frac{1}{\sqrt{2^n}}|x_0\rangle\right)\left(\sqrt{\frac{2^n - 1}{2^n}}\langle\alpha| + \frac{1}{\sqrt{2^n}}\langle x_0|\right)\right] - I$$

The Grover Iterate in 2D

Expressed over the two dimensional subspace:

$$G = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$2|\psi\rangle\langle\psi| - I = 2\left[\left(\sqrt{\frac{2^n-1}{2^n}}|\alpha\rangle + \frac{1}{\sqrt{2^n}}|x_0\rangle\right)\left(\sqrt{\frac{2^n-1}{2^n}}\langle\alpha| + \frac{1}{\sqrt{2^n}}\langle x_0|\right)\right] - I$$

$$2|\psi\rangle\langle\psi - I = \begin{bmatrix} 1 - \frac{1}{2^{n-1}} & \frac{\sqrt{2^n-1}}{2^{n-1}} \\ \frac{\sqrt{2^n-1}}{2^{n-1}} & \frac{1}{2^{n-1}} - 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{bmatrix}$$

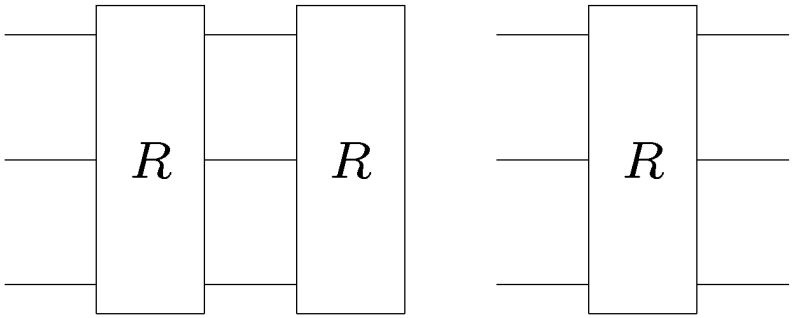
$$\sin\theta = \frac{\sqrt{2^n-1}}{2^{n-1}}$$

$$(2|\psi\rangle\langle\psi| - I)G = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

Grover's iterate is just a rotation in this 2D space

Repeatedly Bang Your Head

Repeated application of the Grover iterate



$$R^k = [(2|\psi\rangle\langle\psi| - I)G]^k = \begin{bmatrix} \cos k\theta & -\sin k\theta \\ \sin k\theta & \cos k\theta \end{bmatrix}$$

Grover's algorithm:

1. start with $|\psi\rangle = \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle$
2. repeatedly apply Grover's iterate to rotate to near $|x_0\rangle$

Repeatedly Bang Your Head

$$R^k = [(2|\psi\rangle\langle\psi| - I)G]^k = \begin{bmatrix} \cos k\theta & -\sin k\theta \\ \sin k\theta & \cos k\theta \end{bmatrix}$$

$$|\psi\rangle = \sqrt{\frac{2^n - 1}{2^n}}|\alpha\rangle + \frac{1}{\sqrt{2^n}}|x_0\rangle$$

Large amplitude in "bad" part of Hilbert space

$$\sin\theta = \frac{\sqrt{2^n - 1}}{2^{n-1}} \quad \text{physicist: } \theta \approx \frac{\sqrt{2^n - 1}}{2^{n-1}} \approx \frac{2}{\sqrt{2^n}}$$

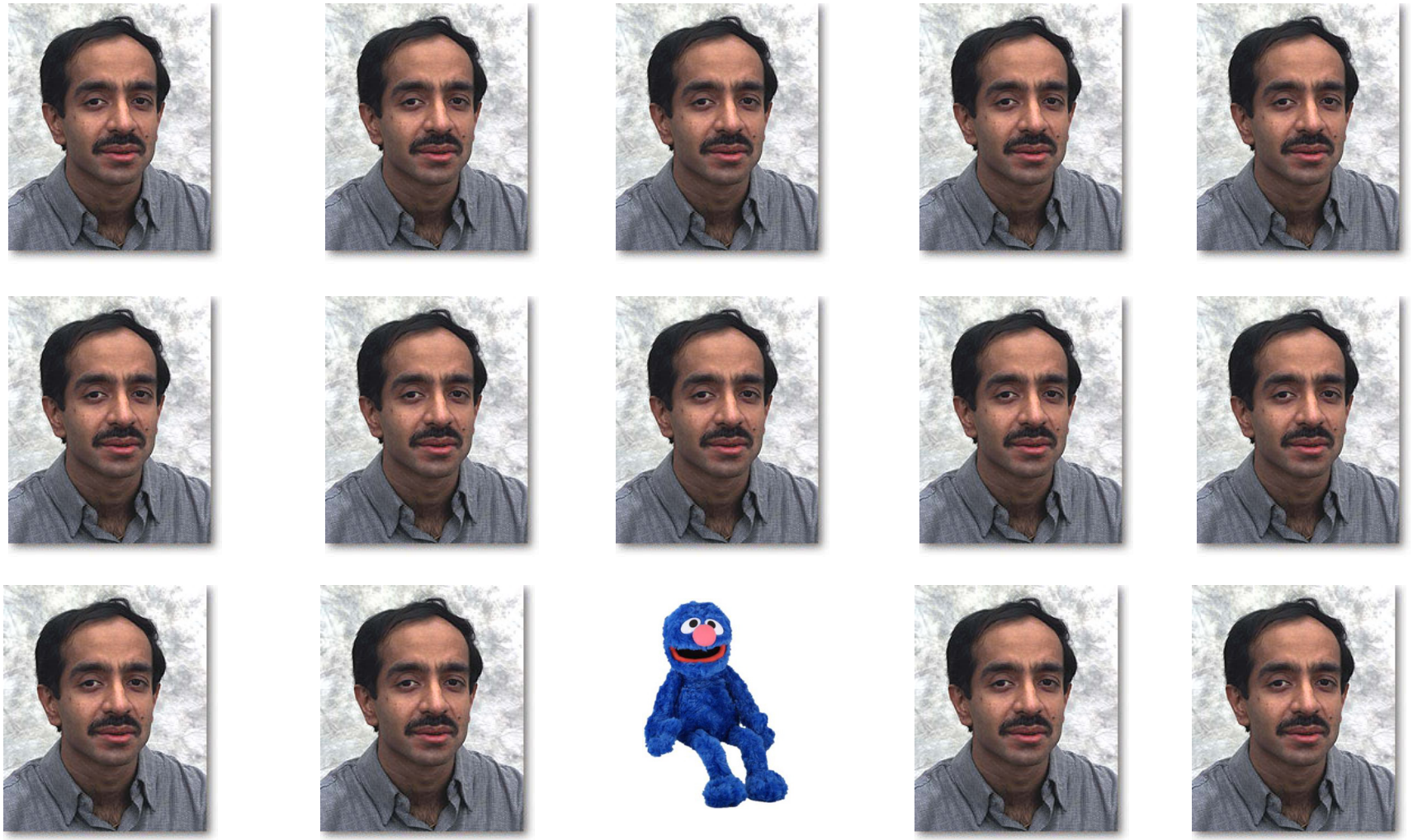
$$k\theta = \frac{\pi}{2} \quad \text{implies} \quad R^k = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \quad k \approx \frac{\pi}{4}\sqrt{2^n}$$

Application of the repeated iterate to initial state rotates it to $|x_0\rangle$

nearly all amplitude in

Gover's Algorithm

We have identified marked item using only $O(\sqrt{2^n})$ queries!



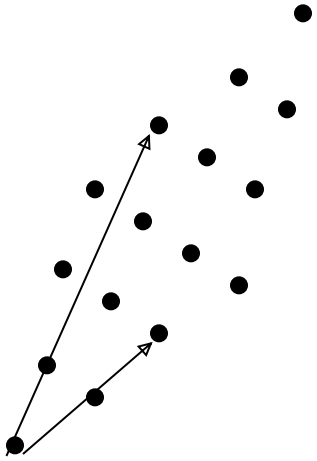
Where Are We Going

What problems are there that quantum computers might speed up?



Graph Isomorphism

Where Are We Going



Certain Shortest Vector in a Lattice Problems

Can breaks Aitai and Dwork cryptosystems

Where Are We Going

$$\begin{bmatrix} 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 1 \end{bmatrix} \qquad \begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 \end{bmatrix}$$

Code Isomorphism

Breaks the McEliece cryptosystem

Where Are We Going

Problems from PPP?

PPP: Polynomial Pigeonhole Principle

Problems which are guaranteed to have a solution due to the pigeon hole principle.

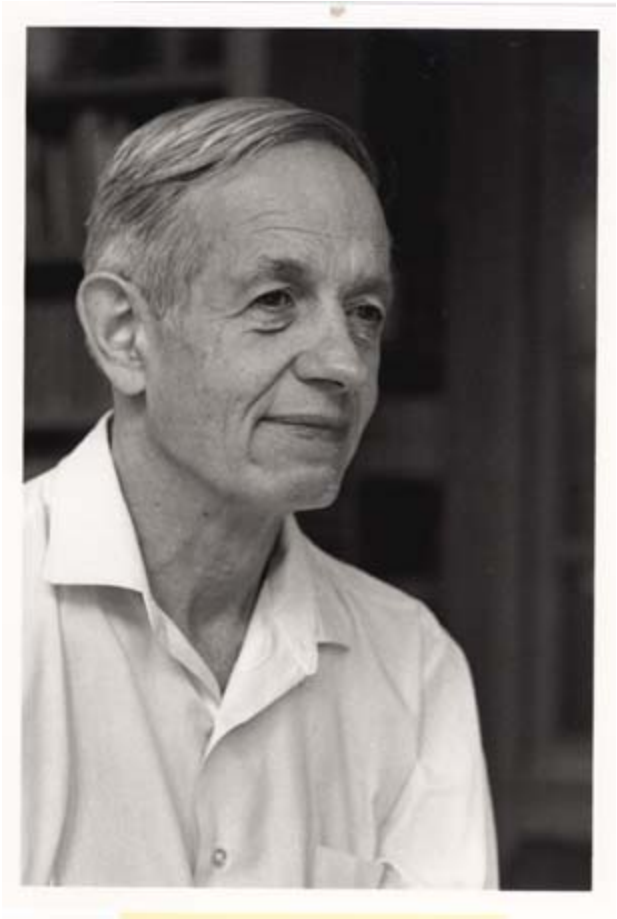
Example:

Given a list $a_1, \ldots, a_n$ of residues mod p and suppose $n > \log_2 p$, find two distinct subsets $S_1, S_2 \subset \{1, 2, \ldots, n\}$ so that

$$\prod_{i \in S_1} a_i = \prod_{i \in S_2} a_j \bmod p$$

Where Are We Going

Finding Nash Equilbrium

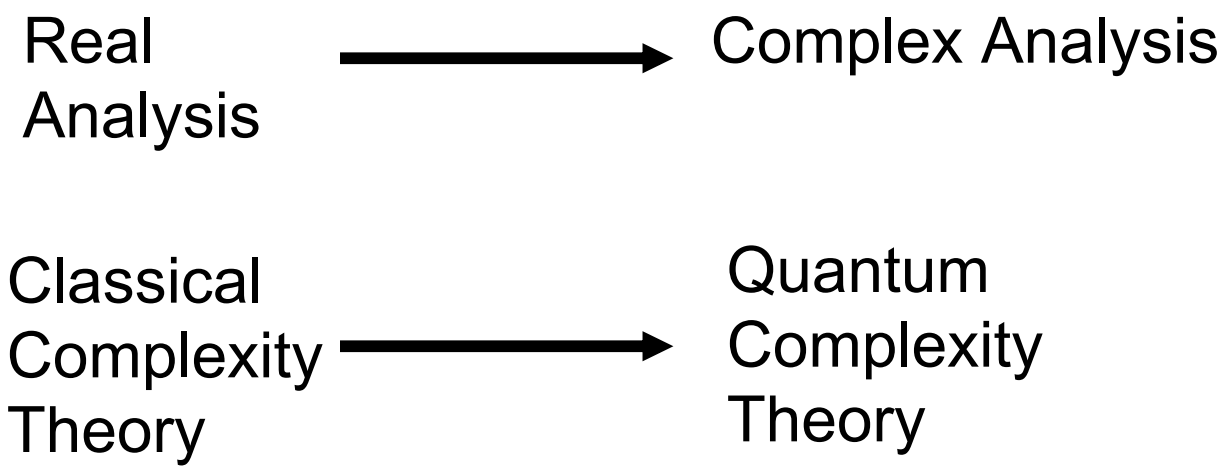


a_n

Where Are We Going

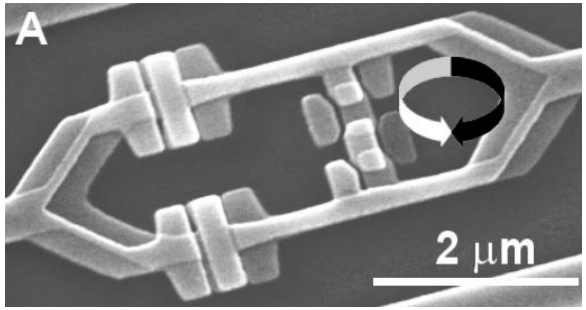
Recently there have been a surge in proof of classical complexity results using quantum methods.

"Assume that computer you had on your lap was actually a quantum computer…then with a lot less work, we can show such and such a complexity result."



SquInT: Quantum Algorithms

188198812920607963838697239461650439807163563379417382700763356422988859715234665485319060606504743045317388011303396716199692321205734031879550656996221305168759307650257059





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